ECVOL-bench: An Open, Identity-Controlled Benchmark for Multimodal Earnings-Call Volatility Prediction

(anonymized working draft for comments — please do not cite or circulate)

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Abstract

A decade of work predicts post-earnings-call stock volatility from the text and audio of quarterly conference calls. We argue that the evaluation substrate this literature shares cannot support its claims: the canonical benchmarks are scored with a zero-day embargo against 30-day targets, with 44–80% of test calls from companies also present in training, on label files inherited unchanged from paper to paper, and under a horizon convention that downstream work misattributes (Table 1, regenerated by one command). We release ECVOL-bench, an open multimodal benchmark over FinCall-Surprise and MAEC that repairs this substrate: reconstructed company identity and call dates for a corpus that ships without them (92.9% of calls; 95.3% of earnings calls; 60/60 manual audit), three target families (level log-realised-volatility, change Δv, HAR-residual) under *both* the calendar-day and trading-day horizon conventions, temporal and ticker-disjoint splits with a 30-trading-day embargo, SHA-256 manifests, strong econometric baselines, and an identity-control suite (linear ticker probes, same-ticker and global content shuffles). Run over modern open-weight text and audio representations, the benchmark returns a consistent answer: text embeddings encode *which company* is speaking (89.5% ticker-probe accuracy; within-ticker transcript shuffles leave performance unchanged), frozen audio content is inert beyond past volatility (its apparent gains survive even a global audio shuffle), and fusion never significantly beats the best unimodal model—under honest controls, apparent content signal in this task family is overwhelmingly company identity plus past volatility. We measure the release timestamp of every call from EDGAR, find that the uniform after-hours anchor every released label set uses mis-places day 0 for 53–72% of calls, and anchor the benchmark's targets on the measured release; re-run the field's runnable models on their own data under these controls (the HTML head loses up to 38% MSE when companies are held out, and the strongest published baseline is same-company history that cannot exist in that condition; nine of seventeen papers ship nothing runnable); re-run the audio ladder on 2025 recordings by bitrate (the audio increment is $\leq$0.09 and does not depend on bitrate); and evaluate the frozen pipeline on 475 post-cutoff calls (ordering preserved; no competitive head degrades). All code, configs, manifests, reconstructed metadata, split files, targets, derived features, and result tables are released.

1 Introduction

Earnings conference calls are one of the few recurring events at which management speaks at length, semi-spontaneously, about a public company—in both *what* is said (the transcript) and *how* it is said (the audio). A line of research beginning with Qin and Yang (2019) and continuing through HTML (Yang et al., 2020), MAEC (Li et al., 2020), KeFVP (Niu et al., 2023), and ECC Analyzer (Cao et al., 2024a) claims that multimodal models over this data predict post-call stock volatility better than classical methods.

Two recent observations cast doubt on how much of that progress is real. First, Yu et al. (2025) ("Same Company, Same Signal") show that transcript representations used across this literature predominantly encode *company identity* rather than content: same-ticker transcripts cluster tightly, and training-free baselines built on a company's own past volatility beat all transcript-based models on their evaluation. Because volatility is strongly autocorrelated and company-specific, a model that learns "which company is this" inherits most of the apparent skill. Second, a 2025 survey of financial NLP (Tatarinov et al., 2025) finds that only $\sim$14% of reported results reproduce exactly, attributing the gap to dead data links, API drift, and contamination.

This paper's premise is that both problems are properties of the *evaluation substrate* the field shares, and that the substrate can be repaired. The lineage above evaluates on a small number of label and split files that are passed from paper to paper essentially unchanged. Those files (§3.1) impose no embargo between training and test windows even though targets run 30 days past each call; they place most test calls' companies in the training set, and no paper reports the ticker-disjoint condition; they carry degenerate label cells; and they encode a horizon convention (calendar days) that the downstream literature describes as trading days. On such a substrate a model cannot be told apart from

a memory of which company it is looking at. Meanwhile the open corpus our audio results rest on, FinCall-Surprise (Shu et al., 2026a), ships with *no* ticker, company, date, or speaker metadata at all—so the identity controls that would expose the confound cannot even be run on it without first reconstructing the identity the dataset omits. A successor release from the same group, MM-FinEval (Shu et al., 2026b), adds ticker and date fields and calendar-day volatility labels, but specifies no embargoed or company-disjoint split, past-volatility baseline, or identity control.

We therefore build and release ECVOL-bench, an open benchmark that supplies what the substrate lacks, and we use it to re-examine the task's central claims. The benchmark is the contribution; the empirical results are the evidence for why it is needed. Concretely, we ask four research questions:

RQ1 Does earnings-call *content* (text and/or audio) add predictive signal for post-call volatility beyond past-volatility persistence (HAR-RV, GARCH)?

RQ2 Is multimodal (audio + text) better than the best unimodal model, after controlling for past volatility?

RQ3 Do the published models of this lineage, re-run on their own data, retain their reported advantage under embargoed, ticker-disjoint evaluation?

RQ4 Do any positive results survive (a) ticker-identity controls, (b) ticker-disjoint splits, and (c) evaluation on post-LLM-knowledge-cutoff calls?

The study was pre-registered so that no single outcome strands the work; the modelling stages, controls, and confirmatory tests were fixed before any model result existed and are recorded in a public decision log, RQ1–RQ4 are all answered here; every number in the paper regenerates from a manifested run.

Contributions.

1. **ECVOL-bench**, an open multimodal volatility benchmark over FinCall-Surprise (primary) and MAEC (secondary): reconstructed identity and dates for FinCall-Surprise (92.9% of calls; 95.3% of earnings calls; 60/60 manual audit), three target families (level v, change Δv, HAR-residual) under both the calendar-day and trading-day horizon conventions, temporal and ticker-disjoint splits with a 30-trading-day embargo, SHA-256 manifests, and a reproducible harness—released in full (§3).

2. **A substrate audit** of the label and split files the prior literature shares: embargo, ticker overlap, label defects, and the horizon convention, regenerable from one command (§3.1, Table 1).

3. **Honest baselines and an identity-control suite**—persistence, EWMA, HAR-RV, GARCH(1,1), and ticker-fixed-effect models for every horizon, split, and target; identity probes; transcript- and audio-shuffle controls; ticker-disjoint evaluation; a gender-confound analysis—and what they show when modern open-weight text and audio representations are run through them on consumer hardware (§6).

4. **Measured call timing**: the earnings-release timestamp of every call from EDGAR 8-K filings (89% of FinCall, 69% of MAEC, 99.8% of Earnings25), showing that the call-date anchor used by every released label set places the reaction session in the pre-window for most calls, with a sensitivity study (§3.4).

5. **A reproduction study** of the runnable models in this lineage on their own data under the new controls, with a code-availability audit of the seventeen papers (one runnable as released, seven partial, nine none; §6.7, Table 8, Table 86).

6. **Post-cutoff evaluation**: the audio ladder re-run on 2025 recordings stratified by bitrate, and the frozen pipeline evaluated on 475 calls held after the knowledge cutoff of every model used (§6.10, §6.11).

2 Related Work

The benchmark lineage. Qin and Yang (2019) introduced text+audio volatility prediction and the 572-call EarningsCall dataset, establishing the 3/7/15/30-day log-realised-volatility target family that the field still uses (defined over *calendar* days in the original, a detail that matters in §3.1). MAEC (Li et al., 2020) scaled this to ∼3,443 sentence-aligned multimodal calls over the S&P 1500. HTML (Yang et al., 2020) introduced a hierarchical transformer with multi-task learning across horizons; VolTAGE (Sawhney et al., 2020a) added graph convolution over inter-stock relations with text/audio fusion; DialogueGAT (Sang and Bao, 2022) modelled speaker/dialogue structure with graph attention. KeFVP (Niu et al., 2023) combined knowledge-enhanced pre-training with a price-conditioned time-series module and is the standard SOTA reference. More recent work moves toward LLMs: ECC Analyzer (Cao et al., 2024a) uses hierarchical LLM summarisation with a RAG "question bank" and wav2vec2 features; RiskLabs (Cao et al., 2024b) fuses calls with news and market series; AT-FinGPT (Liu et al., 2025) is an audio-text LLM. Numeral-aware variants (NumHTML (Yang et al., 2022), NAM/ECNum (Chen et al., 2021a), GNAVol (Shi et al., 2023)) and Q&A-structure models (Ye et al., 2020) target specific aspects of call language. Of these, only a minority release code that still runs; §4 audits which.

The shared substrate. Nearly every paper above evaluates on label and split files derived from Qin and Yang (2019) and redistributed by successors (VolTAGE, KeFVP), rather than recomputing targets from prices. Yu et al. (2025) are the first to report that training-free baselines built on a company's own past volatility beat every transcript model on these files, and release augmented earnings histories that make such baselines easy to build; their follow-up (Yu et al., 2026) times earnings releases from 8-K filings, but for return prediction on a new corpus rather than as an audit of the volatility label sets. No prior work audits the files themselves—embargo, train/test company overlap, label integrity, or the horizon convention—which is the gap §3.1 fills.

Audio and prosody. Chen et al. (2025) report that physics-informed acoustic features explain up to 43.8% of 30-day realised-volatility variance, motivating an audio ladder; the FinAudio benchmark (Cao et al., 2025) cautions that audio-LLMs struggle on long financial audio, informing our preference for frozen pooled features over end-to-end audio-LLMs in v1.

Known pitfalls (the methodological core). Our design is organised around four documented failure modes. *(1) Ticker-identity leakage* (Yu et al., 2025) is the central threat to validity (§1); LLMs also deanonymise entity-neutered call transcripts with near-perfect accuracy (Lopez-Lira et al., 2025), so identity leaks even under masking. We respond with identity-robust targets (Δv, HAR-residual), a ticker-fixed-effect baseline, and an identity-control suite (§5). *(2) LLM lookahead bias*: a 2024-cutoff model "predicting" 2019–2021 outcomes it has effectively seen. A now-active literature offers diagnostics (Gao et al., 2025), cutoff-bounded pretraining (Yan et al., 2026; He et al., 2025), workflow benchmarks that find significant bias in open model families we use (Benhenda, 2026), and memorization filtering (Roy and Roy, 2026); the nearest precedent validates an LLM disagreement measure post-cutoff on ECB press conferences (Collodel, 2025). None evaluates a frozen earnings-call pipeline on post-cutoff calls; we respond with exactly that. *(3) Weak baselines*: much prior work omits persistence, GARCH(1,1), and HAR-RV (Corsi, 2009), which are hard to beat at long horizons; we compute these first and use persistence as the denominator of our headline metric (Gu et al., 2020). *(4) Gender bias in audio features* (Wang et al., 2024; Lin et al., 2024; Mathur et al., 2022): we analyse and report a gender confound rather than debias. We deliberately exclude agentic trading systems and time-series foundation models from v1 scope.

3 The ECVOL Benchmark

We first document the substrate the literature shares (§3.1), then describe what ECVOL-bench supplies in its place: corpora with reconstructed identity (§3.2), targets under both horizon conventions (§3.3), measured call timing (§3.4), leakage-proof splits (§3.5), and the release (§3.6).

3.1 The substrate the literature shares

The EarningsCall (EC) benchmark of Qin and Yang (2019) and the two MAEC releases (2015 and 2016 cohorts; Li et al., 2020) are evaluated, across the lineage in §2, on split and label files that are redistributed from paper to paper rather than recomputed. We fetched the canonical files from the VolTAGE, KeFVP, and HTML repositories, manifested them by SHA-256, and audited four properties. Table 1 (Result Table 5R) is regenerated from those copies by `ecvol audit substrate`; every figure below carries its source file and counts in the released CSV. *(1) Provenance.* The VolTAGE and KeFVP test-split files are byte-identical: the lineage shares one physical label set, never recomputed from prices. *(2) Embargo.* Every train/validation and validation/test boundary has a zero-day gap—on EC, training ends and validation starts on the same date (2017-08-03; 4 training and 14 validation calls on that day)—against targets that run 30 days past each call, so every boundary leaks the target window. *(3) Company overlap.* On the temporal splits the field reports, 80.4% (EC), 44.2% (MAEC-15), and 55.7% (MAEC-16) of test calls come from companies that also appear in training. A temporal split *always* permits this; the finding is not that overlap exists but that no paper in the lineage reports the disjoint condition, so the identity confound of Yu et al. (2025) has never been measured on these benchmarks. *(4) Label integrity and convention.* The single-day log-vol series contain 137 exact-zero cells out of 16,800 (0.82%; the headline averaged series is clean), and KeFVP's MAEC files interleave four binary labels at exactly $\tau \in \{3, 7, 15, 30\}$ among 26 price columns under identical `future_label_N` naming. Finally, the source paper defines its horizons over *calendar* days; most downstream work, and our own design document until its correction, describes them as trading days—at τ=30 the difference is roughly 21 versus 30 sessions, and every published number depends on which one was used. *(5) Labels versus prices.* KeFVP's shipped MAEC test labels key on our own call identifiers; on the 74–77% of calls we can price at $\tau \geq 7$ they correlate only 0.51–0.88 with our calendar-day targets (0.25–0.71 at τ=3, where half the windows hold one session), with mean absolute log-vol differences of 0.15–0.73—so the lineage's labels are not reproducible from prices under any single documented rule.

3.2 Corpora and identity reconstruction

We build on two openly licensed multimodal earnings-call corpora and reserve a third, post-cutoff corpus for the lookahead and clean-audio experiments. **FinCall-**

Benchmark / split	calls (tr/va/te)	overlap	embargo tr→va	embargo va→te	label defect
EC (Qin & Yang)	392 / 56 / 112	80.4%	0 d	0 d	137 / 16,800 zero cells
MAEC-15 (KeFVP files)	535 / 76 / 154	44.2%	0 d	0 d	4 binary cols at τ
MAEC-16 (KeFVP files)	980 / 140 / 280	55.7%	0 d	3 d	4 binary cols at τ
ours: FinCall temporal	1489 / 228 / 483	91.3%	47 d	47 d	—
ours: FinCall ticker-disjoint	1706 / 243 / 484	**0.0%**	n/a	n/a	—
ours: MAEC temporal	1666 / 69 / 529	79.4%	101 d	48 d	—
ours: MAEC ticker-disjoint	1884 / 271 / 533	**0.0%**	n/a	n/a	—
ours: Earnings25 ticker-disjoint	333 / 47 / 95	**0.0%**	n/a	n/a	—

Table 1: Substrate audit of the shared benchmark files (Result Table 5R, regenerated by `ecvol audit substrate` from SHA-256-manifested copies of the VolTAGE and KeFVP repositories). The VolTAGE and KeFVP split files are byte-identical. Overlap is a property every temporal split permits, ours included; the point is that no paper in the lineage reports the disjoint condition. The zero-day embargos leak 30-day target windows across every boundary. Shipped MAEC labels join our calendar-day, call-date-anchored targets on 77–79% of test calls at $\tau \geq 7$ and correlate 0.52–0.87 with them (0.25–0.66 at τ=3).

Surprise (Shu et al., 2026a) (primary; Apache-2.0) contains 2,688 calls spanning 2019–2021 with full MP3 audio, JSON transcripts, and slides. **MAEC** (Li et al., 2020) (secondary; CC-BY-SA-4.0) contains ∼3,443 calls over 2015–2018 (S&P 1500) with sentence-aligned audio features and transcripts but no raw audio. **Earnings25** (Jiang et al., 2026) (CC-BY-4.0; 514 S&P 500 calls, 498 h of audio, held July–December 2025 with 97% in October–December) is post-cutoff for every model we use and carries webcast MP3 audio at 64 kbps for 281 calls and ≤24 kbps for 232 (one at 32 kbps); it is the post-cutoff corpus (§5). We admit a call when its company resolves to an SEC ticker that was an S&P 500 constituent on the release date (476 of 514; the rest are name-collision look-alikes such as Grainger PLC or Paramount Group, reason-coded), and 475 join prices.

FinCall-Surprise ships with four fields per record and *no* ticker, company, CIK, date, time, fiscal period, or speaker names. We reconstruct identity and date from transcript content and slide metadata, cross-check against price data, and release the resulting table together with an 80-row manual override list: the reconstruction resolves 2,496/2,688 calls (92.9% overall; 95.3% of the 2,499 earnings-type calls) and passes a 60/60 manual audit. This table is what makes the identity controls of §5 runnable on the corpus at all.

3.3 Price data and targets

Daily adjusted-close OHLCV is obtained from a free, no-account source and cross-checked against Tiingo on a 50-ticker random sample (47 pass, three flagged and documented, minimum adjusted-return correlation 0.976); delisted or unmatched tickers produce an explicit coverage report rather than being silently dropped. After price joining, FinCall-Surprise yields 2,433 calls with at least one target (97.5% join rate) and MAEC yields 2,688 (78.6%; a documented shortfall driven by 2015–2018 delistings), for a combined modelling set of 5,121 calls.

For each call we define returns $r_t = (P_t - P_{t-1})/P_{t-1}$ relative to day $t = 0$, the last trading day before the call's information is public (§3.4). The primary target is log realised volatility over horizon $\tau \in \{3, 7, 15, 30\}$,

$$v_{\text{post}}(\tau) = \ln\left(\sqrt{\tfrac{1}{\tau}\textstyle\sum_{t=1}^{\tau}(r_t - \bar{r})^2}\right), \qquad (1)$$

with pre-call volatility $v_{\text{pre}}(\tau)$ defined over the sessions ending at day 0. We ship every target under *both* horizon conventions. Under the **trading-day** convention the window is sessions $1..\tau$ ($n = \tau$). Under the **calendar-day** convention of Qin and Yang (2019), which the published leaderboard inherits, the window is every session dated within τ calendar days of day 0, so $n \approx 0.7\tau$ and varies with the weekday and holidays (FinCall: $n = 2.4/4.9/10.7/21.0$ sessions on average at $\tau = 3/7/15/30$). In both cases the variance is taken over the n sessions actually present, and a window holding fewer than two sessions is excluded with a reason code rather than reported as a one-return "volatility"—which removes **48.8%** of FinCall's joined calls at τ=3 calendar days (every Thursday or Friday call). Each convention is a separate file with an identical schema, a `convention` column and the per-row session counts; the trading-day file is the validated set every result in §6 uses, and every result table carries the convention as a column (§6.6). We add two identity-robust variants: the **change** $\Delta v(\tau) = v_{\text{post}}(\tau) - v_{\text{pre}}(\tau)$, which subtracts the company's own level, and the **HAR-residual** $v_{\text{post}}(\tau) - \widehat{\text{HAR}}(\tau)$, deviation from a train-only HAR-RV (Corsi, 2009) forecast. Insufficient post-call history and zero-variance windows yield `NaN` targets excluded with a reason code; targets are unit-tested against hand-computed values.

3.4 Call timing

Fewer than 4% of FinCall-Surprise transcripts mention a clock time, and 32.4% of recovered *dates* come from slide-PDF creation stamps, so no call in either corpus carries a measured time of day. We therefore measure the information boundary per call from EDGAR: the acceptance time of the earliest 8-K carrying Item 2.02 (the earnings release) within three days before to one day after the call date, taken from the SEC submissions API and cached. Where no such filing exists we fall back, in order, to Earnings25's call-start time, to the before/after-market attribute of Yu et al. (2025), and finally to the after-hours assumption (calls assumed to occur after market close; t=1 is the next trading session), with the tier recorded per call. Coverage by the EDGAR tier is 89.3% for FinCall-Surprise, 68.6% for MAEC, and 99.8% for Earnings25. A release before the

open makes the call date itself day 1; intraday releases are treated as after-hours and flagged. **Every result in §6 uses these measured anchors.** They matter: against the call-date anchor that every released label set in this literature uses (calls assumed after-hours), **day 0 moves for 53% of FinCall, 56% of MAEC and 72% of Earnings25 calls**, almost all one session earlier, because releases before the open are the norm. Under the call-date anchor the reaction session therefore sits in the *pre* window for most of the corpus, and the published targets share that error. We ship the call-date targets beside the primary ones (`targets_assumed`) so results remain comparable with prior label sets, with a sensitivity table (`timing_sensitivity.csv`): on the measured anchor the persistence MSE is up to 11% higher on FinCall, 25% on MAEC and 27% on Earnings25, because the reaction session now enters the post window, while the HAR-RV R^2_{OOS} moves by -0.07 to $+0.13$. The learned ridge heads are more anchor-sensitive than HAR-RV (§6.11).

3.5 Leakage-proof splits

Three split families are generated once as committed CSVs and guarded by leakage assertions in CI. *Temporal*: train < validation < test in strict calendar order (∼70/10/20) with a **30-trading-day embargo** between segments so that no 30-day target window crosses a boundary—the constraint the shared substrate lacks (§3.1). *Ticker-disjoint*: companies in validation/test never appear in training (grouped, seeded), the headline identity-control split; by construction its test-call company overlap is 0.0%, against 91.3% (FinCall-Surprise) and 78.9% (MAEC) on our own temporal splits. *Combined*: both constraints simultaneously, reported as the hardest robustness row. An information rule is codified and enforced: every feature function receives an `as_of` timestamp, and the harness fails any feature that reads price data after it. FinCall-Surprise spans COVID; per-year breakdowns are mandatory and a Feb–Apr 2020 exclusion is reported as an exploratory sensitivity.

3.6 Release

ECVOL-bench is released in full: the identity/date reconstruction table and override list, split CSVs, targets under both conventions, the derived-feature archives of §4, the evaluation harness and baselines, the substrate-audit report, and every result table. FinCall-Surprise-derived artifacts are Apache-2.0; MAEC-derived artifacts inherit CC-BY-SA-4.0. Raw audio and transcripts from sources with unclear licenses are never redistributed (a *scripts-not-data* policy).

4 Models

We organise our own models as a ladder (Stages 0–4; Figure 1). Two invariants make the comparison honest. First, from Stage 2 upward every learned model includes past-volatility covariates (v_{pre} at all horizons plus the HAR-RV forecast) and is *also* reported with them ablated, baking the persistence control into every comparison. Second, given the small-data regime (10^3–10^4 samples) we use frozen representations with light heads (ridge, shallow MLP, LightGBM) and per-horizon models by default; fine-tuning is gated to a conditional later stage and a shared multi-task head is an ablation, not the headline. Because the shallow-MLP heads proved seed-unstable (across-seed R^2_{OOS} std up to ≈3.4; see Limitations), the consolidated grid (Table 6) uses one pre-registered deterministic ridge/structural canonical model per stage; MLP heads appear in the appendix only.

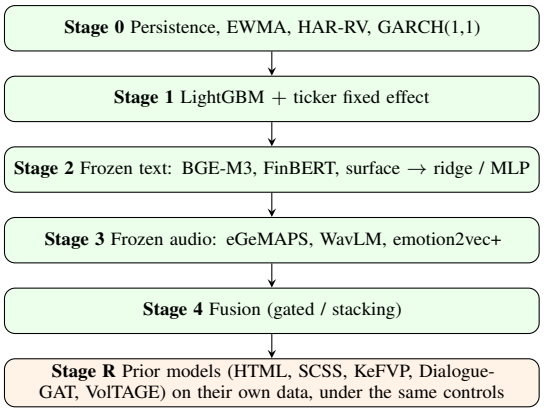


Figure 1: The modelling ladder. From Stage 2 up, every model also carries past-volatility covariates (and is reported with them ablated). Every stage runs through the identity-control suite. Green = complete, orange = pending.

Stage 0 — econometric baselines. Persistence ($\hat{v} = v_{\text{pre}}$), EWMA (RiskMetrics λ=0.94), HAR-RV (OLS on daily/weekly/monthly realised-variance factors, Corsi, 2009), and GARCH(1,1) (per-call fit, `arch`). These establish the floor and the denominator of the headline metric.

Stage 1 — tabular identity baseline. LightGBM over past-volatility features, trivial metadata (transcript length, Q&A turn count where available), and a **ticker fixed effect**. This is the in-house "Same-Company-Same-Signal" control: it measures how much apparent multimodal performance is identity plus metadata.

Stage 2 — frozen text content. We segment each transcript into prepared-remarks vs. Q&A and speaker turns, then extract three feature blocks: BGE-M3 (BAAI, 2024) section-pooled sentence embeddings (1024-d), FinBERT (ProsusAI, 2019) sentiment aggregates per scope×role, and surface statistics. Heads are ridge (with the α grid selected on validation, *no* retrain on train+val) and a shallow MLP (1×256, L2 + early stopping, 5 seeds). Features use fp32 with content-hash

caching for bit-identical re-extraction.

Stage 3 — frozen audio content. Audio is QC'd and resampled to 16 kHz mono FLAC. We extract eGeMAPSv02 functionals (88-d, openSMILE (Eyben et al., 2010), CPU, interpretable), then WavLM-Large (Chen et al., 2021b) and emotion2vec+ (Ma et al., 2023) embeddings over 30 s non-overlapping windows, mean-pooled to a per-call vector (2,671 calls, all extracted on one consumer GPU with resumable content-hash caches), fed to the same heads. We use per-call pooling only (no per-sentence alignment), and report a gender-confound analysis.

Stage 4 — fusion. Gated fusion (per-modality train-fit PCA + norm gates → shallow MLP) and late-fusion stacking (a meta-ridge over the text, audio, and Stage-1 GBDT base predictions, fit on validation only) over frozen modality embeddings, to test whether multimodal beats the best unimodal model (RQ2). Cross-attention is omitted: with one pooled vector per modality, attention over 2–3 tokens reduces to gated fusion. All fusion heads are <5M parameters.

Reproduced prior models. Of the seventeen papers in the lineage of §2 that we probe (Table 86), one releases code that runs as released: Same-Company-Same-Signal (Yu et al., 2025). Seven ship code that needs repair or missing inputs before it starts—HTML (Yang et al., 2020), KeFVP (Niu et al., 2023), DialogueGAT (Sang and Bao, 2022), Sawhney et al. (2020b) and VolTAGE (Sawhney et al., 2020a) among them. We re-run each of the five on the slate *on its own original dataset*, with the authors' code wherever it can be made to start (every patch listed in a released fidelity ledger) and a line-by-line port where it cannot; we first compare against the published-split number, then evaluate the same model under an embargoed and a ticker-disjoint split of that dataset, so that "the model fails" cannot be confused with "the model does not transfer" (RQ3). The remaining nine—among them the strongest recent positive claims (Cao et al., 2024a; Chen et al., 2025)—have no code, empty or unpushed repositories, closed-API dependencies, or a stated inability to run; we tabulate each with the evidence (Table 8; Table 86 in Appendix A).

5 Experimental Protocol

Metrics. We report MSE on log-RV per horizon (literature comparability); R^2_{OOS} vs. persistence (Gu et al., 2020; $1 - \sum(y-\hat{y})^2 / \sum(y-\hat{y}_{\text{persist}})^2$), the headline metric, which is not gameable by identity memorisation the way raw MSE is; MAE; and cross-sectional Spearman rank correlation per calendar quarter. All metrics are reported on all three targets (v, Δv, HAR-residual) × all splits, and carry the horizon convention as a column.

Statistical rigor. Every learned model is run with ≥5 seeds (mean ± std). We use Diebold–Mariano tests (Diebold and Mariano, 1995) (HLN small-sample correction) on per-call squared-error differentials vs. persistence and vs. the best preceding stage; cluster-bootstrap confidence intervals (1,000 resamples) clustered by ticker and by calendar quarter; and Holm-corrected p-values across horizons for confirmatory claims. Confirmatory vs. exploratory analyses are labelled in advance: confirmatory analyses are Stage-k vs. Stage-1 DM tests on Δv at each τ, Stage 4 vs. best unimodal, the temporal-vs-ticker-disjoint gap per stage, and—for each reproduced prior model—the published-split vs. leakage-proof-split gap; all per-year/regime, COVID-exclusion, prepared-vs-Q&A, call-length, and gender-confound analyses are exploratory.

Identity-control suite. The signature diagnostic of this benchmark probes whether content models read content or identity, via four controls. *(1) Ticker-only model*: predict the per-ticker target mean (global mean for unseen tickers); if a content model matches it on level-v, that model reads identity. *(2) Same-ticker transcript shuffle*: replace each test call's transcript with a different call from the *same* ticker; if performance is unchanged the model reads identity, not content, whereas a *global* shuffle (any other call) isolates pure overfitting. *(3) Identity probe*: a linear probe predicting the ticker from frozen call embeddings; high accuracy demonstrates identity is encoded. *(4) Ticker-disjoint split* results reported beside temporal-split results in every table. The suite runs unchanged on the reproduced prior models (§4).

Post-cutoff experiments. Earnings25 (§3.2) supports two experiments that the 2019–2021 corpora cannot. *Clean audio*: the frozen Stage-3 audio ladder (same extractors, same heads, same controls) is re-run on source-quality 2025 recordings, to test whether the audio-inert finding of §6 is a property of the task or of FinCall-Surprise's uniform 40 kbps transcodes. *Lookahead*: the entire frozen pipeline—no retraining, no post-hoc threshold changes (rule fixed in advance)—is evaluated on calls held after the knowledge cutoff of every model used; disproportionate degradation relative to the 2019–2021 results would be contamination evidence (§6.10, §6.11).

Reproducibility. All result tables regenerate byte-identically from committed CSV artifacts via `ecvol report`, guarded by a CI test. Every result CSV carries a committed run manifest—the resolved command config and its hash, git SHA, seeds, environment fingerprint, and the file's SHA-256—and both `ecvol report` and CI refuse a result file whose bytes match no manifest; one committed config per result-producing command is the record of how it was run. Every data

file carries a SHA-256 manifest entry; targets are unit-tested against hand-computed values; and the splits are guarded by leakage assertions in CI. Every result that predated this mechanism has since been re-run by its producing command and reproduced its committed file byte for byte, five-seed MLP heads included. `REPRODUCE.md` lists the one command per table, and `ecvol audit predictions` applies the identity controls to any prediction file (company overlap and embargo of the split, ticker-only and persistence floors, same- and other-company prediction swaps, the company share of prediction variance, company-clustered significance), so the diagnosis of §6.9 can be run on a model without adopting this pipeline; a clean-machine regeneration of Table 2 is part of the release checklist.

6 Results

We report results factually and defer interpretation to §7. Curated headline tables appear here on the identity-robust Δv target for FinCall-Surprise under the trading-day convention; the complete grid (both corpora, level-v and Δv, R^2_{OOS} and MSE, all splits and horizons) is in Appendix A. Stars denote Diebold–Mariano significance ($p < 0.05$): in Table 2 vs. persistence, and in Tables 3 and 5 vs. the Stage-1 ticker-FE GBDT; in the consolidated grid (Table 6) markers are Holm-corrected across horizons and signed ($*$ better / † worse than Stage-1).

6.1 Baselines (Stages 0–1)

Table 2 shows the econometric floor. HAR-RV is strong at short and medium horizons on both splits, confirming RQ1's premise that past-volatility baselines are hard to beat. The one striking failure is the FinCall *temporal* split at τ=30, where every model falls below persistence (HAR-RV $R^2_{\text{OOS}} = -0.287$): the training window is COVID-inflated (Feb–May 2020) while the test window is calm late 2021, so a long-horizon level forecast trained on the crisis regime is mis-calibrated. The same models pass comfortably on the ticker-disjoint split (HAR-RV $R^2_{\text{OOS}} = +0.229$ at τ=30) and on MAEC, so we treat this as a documented regime exception rather than a target bug; per-year breakdowns are mandatory in the appendix.

6.2 Text content (Stage 2)

Table 3 shows frozen-text heads against the Stage-1 baseline on Δv. The result is a clean negative: *text content does not cleanly beat the identity/past-volatility baselines*. On the FinCall temporal split the text ridge head is near zero and collapses at τ=30; the MLP heads overfit badly. On the ticker-disjoint split the text ridge head is weak (R^2_{OOS} between -0.04 and $+0.17$) while the past-vol-only head is consistently strong ($R^2_{\text{OOS}} \approx 0.19$–$0.45$), so any apparent text gain is not separable from what past volatility already provides. MAEC shows the same pattern at smaller magnitude (Appendix A).

Model	τ=3	τ=7	τ=15	τ=30
Temporal split (n=483)				
Persistence	0.000	0.000	0.000	0.000
EWMA	0.465*	0.389*	0.259*	0.019
HAR-RV	0.444*	0.372*	0.138*	-0.285*
GARCH(1,1)	0.472*	0.463*	0.298*	0.023
GBDT (ticker FE)	0.340*	0.249*	-0.126	-0.948*
Ticker-disjoint split (n=484)				
Persistence	0.000	0.000	0.000	0.000
EWMA	0.520*	0.379*	0.142*	0.112*
HAR-RV	0.570*	0.491*	0.349*	0.345*
GARCH(1,1)	0.561*	0.484*	0.334*	0.298*
GBDT (ticker FE)	0.494*	0.440*	0.292*	0.237*

Table 2: Baselines. R^2_{OOS} vs. persistence on the Δv target, FinCall-Surprise (test). HAR-RV is strong except on the COVID-regime temporal τ=30 cell. $*$: Diebold–Mariano significant vs. persistence ($p < 0.05$).

Model	τ=3	τ=7	τ=15	τ=30
Temporal split (n=483)				
Ridge (text)	0.182*	0.202	0.044*	-0.133*
Ridge (past-vol)	0.412*	0.314*	0.094*	0.113*
Ridge (text+vol)	0.333	0.285	0.080*	-0.077*
MLP (text)	0.013*	0.069*	-0.253	-1.202
Ticker-disjoint split (n=484)				
Ridge (text)	0.195*	0.217*	0.195*	0.169
Ridge (past-vol)	0.552*	0.469	0.345	0.318*
Ridge (text+vol)	0.399*	0.375	0.256	0.301
MLP (past-vol)	0.577*	0.508*	0.381*	0.356*

Table 3: Text content (Stage 2). R^2_{OOS} vs. persistence on the Δv target, FinCall-Surprise (test). The text-only head does not separate from the past-volatility head; adding text to past-vol does not help. $*$: Diebold–Mariano significant vs. the Stage-1 ticker-FE GBDT ($p < 0.05$). Full head set in Appendix A.

6.3 Identity controls (Stage 2 diagnostic)

Table 4 explains the negative result. A linear probe recovers the ticker from frozen call embeddings at 89.5% on FinCall (319× chance) and 53.9% on MAEC (487× chance): identity is heavily encoded. The transcript shuffle is decisive—replacing each test call's transcript with a *different call from the same ticker* barely changes R^2_{OOS}, whereas a global shuffle degrades it. The heads are reading *which company* is speaking, not *what* is said. This is the in-corpus confirmation of Yu et al. (2025) and the reason RQ4 is central—and it is only measurable because the benchmark reconstructed the identity the corpus omits (§3.2).

Corpus	Identity probe (linear) Accuracy	× chance
FinCall (n=2656, 357 tickers)	89.5%	319×
MAEC (n=3132, 902 tickers)	53.9%	487×

Condition	τ=3	τ=7	τ=15	τ=30
Real transcript	0.182	0.202	0.044	-0.133
Within-ticker shuffle	0.174	0.194	0.047	-0.110
Global shuffle	0.145	0.110	0.008	-0.152

Table 4: Identity controls. Top: a linear probe predicts the ticker from frozen BGE-M3 call embeddings far above chance, demonstrating identity is heavily encoded. Bottom: R^2_{OOS} of the text ridge head on the Δv target (FinCall, temporal split, test, n=483). Replacing each test call's transcript with a *different call from the same ticker* (within-ticker shuffle) barely changes performance, whereas a global shuffle degrades it—evidence the head reads ticker identity rather than call content.

6.4 Audio content (Stage 3)

Table 5 extends the same design to audio. Audio-*only* ridge heads (eGeMAPS, WavLM-Large, emotion2vec+) sit at R^2_{OOS} zero on the temporal and ticker-disjoint splits (-0.06 to $+0.03$), and their MLP counterparts fall well below persistence at long horizons (to -1.3 at τ=30): frozen audio content alone carries no measurable Δv signal. Adding past-volatility covariates reproduces the Stage-2 pattern—the only DM-significant wins over Stage-1 are past-vol-augmented heads on the ticker-disjoint split. The audio-shuffle control (bottom block) is decisive: replacing each test call's audio with a random *other company's* call leaves most of those gains intact (τ=3: 0.337→0.318; τ=30: 0.212→0.100), so the audio features are inert and past volatility carries the signal. Identity is nonetheless encoded: a linear probe recovers the ticker from per-call WavLM embeddings at 76.3% (271× chance; emotion2vec+ 29.4%, 105×). An exploratory gender-confound check (eGeMAPS-F0 dominant-speaker proxy, 100% coverage) finds near-identical per-group test MSE (0.241 low-pitch vs. 0.242 high-pitch) and a 0.03 correlation between speaker F0 and squared error: with inert audio features, no large gender disparity is detectable (see the Ethics Statement for caveats). All of this rests on FinCall-Surprise's 40 kbps source audio; §6.10 is the test of whether it transfers.

6.5 Fusion and the consolidated grid (Stage 4)

Gated fusion and late-fusion stacking over all frozen modalities answer RQ2 in the negative: *multimodal is not better than the best unimodal model*. Only one of 48 Δv test cells beats Stages 1–3 DM-significantly (gated fusion with past-vol covariates, ticker-disjoint τ=7, R^2_{OOS}= + 0.019—negligible), and the best fusion cell overall (stacking, ticker-disjoint τ=3, R^2_{OOS}=0.436) is not significant vs. Stage-1 (p=0.13). Table 6 consolidates the ladder with one pre-registered canonical model per stage and Holm-corrected confirmatory tests. Two facts summarise it. On the regime-honest *temporal* split, no content stage beats the econometric floor at any horizon; where a content model is Holm-significantly better than Stage-1 (fusion at τ=30) it merely beats a collapsed GBDT while remaining below persistence. On the *ticker-disjoint* split the favorable cells are exactly the past-vol-augmented ones that the shuffle controls (Tables 4 and 5) show to be identity/past-vol-confounded. The per-year breakdown sharpens the regime point: the FinCall temporal test year is entirely 2021, and there every content model underperforms persistence on level-v (HAR -0.29, GBDT -0.82, text -1.09, audio -1.66, fusion -1.02).

Model	τ=3	τ=7	τ=15	τ=30
Temporal split (n=483)				
Ridge (eGeMAPS)	0.163*	0.156*	0.059*	0.079*
Ridge (WavLM)	0.156*	0.148*	-0.027	-0.008*
Ridge (emotion2vec+)	0.155*	0.129*	0.022*	0.027*
Ridge (WavLM+past-vol)	0.357	0.262	-0.044	0.052*
Ticker-disjoint split (n=484)				
Ridge (eGeMAPS)	0.203*	0.119*	0.137*	0.044*
Ridge (WavLM)	0.178*	0.192*	0.131*	0.026*
Ridge (emotion2vec+)	0.181*	0.185*	0.122*	0.047*
Ridge (WavLM+past-vol)	0.456	0.390	0.267	0.222
Audio-shuffle control of WavLM+past-vol (ticker-disjoint)				
Real audio	0.456	0.390	0.267	0.222
Within-ticker shuffle	0.453	0.373	0.229	0.152
Global shuffle	0.453	0.323	0.164	0.099

Table 5: Audio content (Stage 3). R^2_{OOS} vs. persistence on the Δv target, FinCall-Surprise (test). Audio-only heads sit at zero everywhere; the apparent gains of the past-vol-augmented head survive replacing every test call's audio with a *random other company's* call (global shuffle, bottom block), so past volatility, not audio content, carries them. $*$: Diebold–Mariano significant vs. the Stage-1 ticker-FE GBDT ($p < 0.05$; read direction from the value). Full head set in Appendix A.

6.6 Horizon convention: calendar vs. trading days

Table 7 reports the delta between the trading-day targets used above and the calendar-day targets of Qin and Yang (2019). Three facts matter for anyone comparing to the published leaderboard. First, the calendar window holds about 0.7τ sessions, and at τ=3 it holds at most one session for any Thursday or Friday call: 48.8% of FinCall's calls have no defensible τ=3 calendar-day target and are excluded under our rules, whereas the released label sets carry a value for every call. Second, the two conventions are far from interchangeable at short horizons (correlation of v_{post} 0.75 at τ=3, 0.93–0.96 beyond), and the persistence MSE—the leader-

Stage (canonical model)	τ=3	τ=7	τ=15	τ=30
Temporal split (n=483)				
S0 HAR-RV	0.444	0.372	0.138	-0.285
S1 GBDT (ticker FE)	0.340	0.249	-0.126	-0.948
S2 Text (ridge+vol)	0.333	0.285	0.080*	-0.077*
S3 Audio (ridge+vol)	0.353	0.268	0.033*	0.050*
S4 Fusion (stack)	0.406*	0.347*	0.066*	0.100*
Ticker-disjoint split (n=484)				
S0 HAR-RV	0.570	0.491	0.349	0.345
S1 GBDT (ticker FE)	0.494	0.440	0.292	0.237
S2 Text (ridge+vol)	0.399†	0.375	0.256	0.301
S3 Audio (ridge+vol)	0.442	0.364	0.241	0.217
S4 Fusion (stack)	0.531	0.465	0.340	0.396*

Table 6: Consolidated grid (Stages 0–4), pre-registered canonical model per stage. R^2_{OOS} vs. persistence on the Δv target, FinCall-Surprise (test); persistence is 0 by construction. Significance is the Holm-corrected confirmatory family (Stage-k vs. Stage-1 DM across horizons): $*$ = significantly *better* than Stage-1, † = significantly *worse* ($p < 0.05$). No content stage beats the econometric floor on the temporal split (the fusion τ=30 star beats only the collapsed GBDT, not persistence); the ticker-disjoint gains are past-vol/identity-confounded (Table 5, bottom block). Canonical models are deterministic ridge/structural heads; seed-unstable MLP heads are reported in Appendix A only.

	τ=3	τ=7	
Targets (FinCall, calls ok under both conventions)			
Sessions in calendar window (mean)	2.4	4.9	
Calls with $\geq$2 sessions (of 2,433)	1,398	2,433	
corr($v^{\text{trad}}_{\text{post}}, v^{\text{cal}}_{\text{post}}$)	0.784	0.942	
corr($\Delta v^{\text{trad}}, \Delta v^{\text{cal}}$)	0.751	0.887	
Persistence MSE, level-v, temporal test (trading / calendar)			
FinCall	1.270 / 2.071	0.448 / 0.612	0.211
MAEC	1.774 / 2.553	0.712 / 0.986	0.356
HAR-RV R^2_{OOS} vs. persistence, level-v, temporal test (trading / calendar)			
FinCall	0.444 / 0.419	0.372 / 0.402	0.138
MAEC	0.603 / 0.562	0.486 / 0.502	0.321

Table 7: Horizon convention delta. The calendar-day windows of Qin and Yang (2019) hold $\sim$0.7τ sessions; at τ=3 43% of the calls have at most one post-call session and are excluded. Persistence MSE (the leaderboard's metric) rises by 44–63% at τ=3 under the calendar convention, so published MSE values are not comparable across conventions; the FinCall temporal τ=30 deficit of HAR-RV against persistence (the COVID-regime cell, §6) holds under both conventions (-0.285 / -0.053). Full model $\times$ horizon views under both conventions: `data/results/result_table_1.tex`.

board's metric—roughly doubles at τ=3 under calendar days because the target is averaged over two returns instead of three. A published MSE therefore carries no information about model quality unless the convention is stated. Third, the one cell where our baselines fail the pre-registered sanity gate (HAR-RV below persistence at τ=30 on FinCall's temporal split, the COVID-regime cell) becomes a marginal pass under the calendar convention (R^2_{OOS} $-0.287 \rightarrow +0.017$), because a 21-session window straddles less of the regime shift than a 30-session one. The Stage-0/1 grid is shipped under both conventions (`result_table_1`, convention column); the Stage-2–4 content models are evaluated under the trading-day convention only, since the delta above is a property of the target, not of the head.

6.7 Prior models under the new controls (Stage R)

Table 8 reports the first entry of the reproduction study: the HTML sentence-level transformer with its multi-task loss and published hyper-parameters (Yang et al., 2020), run over our frozen BGE-M3 chunk sequences (a documented encoder substitution; the head and protocol are the paper's) on EC. Three things follow. First, the published number is only partly reproduced: at τ=3 the head beats it (0.77 vs. 1.18), but the published τ=15 and τ=30 values (0.153, 0.133) are *below the persistence floor of the shipped labels* (0.340, 0.202) and no configuration we ran approaches them—the head is worse than persistence at τ=30 in every condition, as every model in §6 is on FinCall. Second, label source barely matters (0.77/0.44/0.29/0.22 vs. 0.76/0.42/0.29/0.21), so the convention and anchor issues of §3.1 do not explain the shortfall. Third, the ticker-disjoint split raises the head's MSE by 2% / 20% / 38% / 36% over the published split with identical labels, collapsing its R^2_{OOS} against persistence at τ=15 from 0.20 to 0.06, whereas the 30-session-embargoed temporal split costs 2% / 7% / $-$2% / 24%—at the horizons where the head has any signal, the company-overlap confound, not the embargo, is what the published protocol leaves unmeasured. The second entry is Same-Company-Same-Signal (Yu et al., 2025) on its own DEC data: its training-free PEV and STPEV baselines reproduce to three decimals from the shipped file (2023 quarterly averages 0.240 / 0.253 / 0.227 / 0.246 for STPEV against the published 0.239 / 0.253 / 0.227 / 0.246), which confirms the paper's own point in the strongest form: STPEV is same-company history and has *no definition* under a ticker-disjoint evaluation, so the best model on DEC is one that cannot be run in the condition our controls require. Of seventeen papers in the lineage, one ships code that runs as released, seven ship code that needs repair or missing inputs before it starts, and nine ship nothing usable (Table 86, Appendix A). The next subsection goes further for the five with usable code.

6.8 Reproductions with the authors' own code

Table 9 re-runs the five papers with usable code on their own data, with the authors' code wherever it runs and a line-by-line port where it cannot (§4); the fidelity ledger released with the code records every patch and substi-

Condition (EC)	model	τ=3	τ=7	τ=15	τ=30
paper, published split, published labels	HTML (text, WWM-BERT)	1.175	0.372	0.153	0.133
published split, published labels	HTML head (BGE-M3)	0.769	0.436	0.285	0.223
	persistence	1.485	0.570	0.340	0.202
published split, our labels	HTML head (BGE-M3)	0.760	0.420	0.285	0.214
	persistence	1.826	0.590	0.333	0.183
embargoed temporal, our labels	HTML head (BGE-M3)	0.778	0.448	0.278	0.265
	persistence	1.948	0.629	0.353	0.192
ticker-disjoint, our labels	HTML head (BGE-M3)	**0.773**	**0.503**	**0.394**	**0.291**
	persistence	1.670	0.728	0.418	0.213

Table 8: Reproduced prior model under leakage-proof evaluation (Result Table 6R, first entry). The HTML head reproduces the paper's ordering against price baselines at τ=3 (and beats the published number) but not the published τ=15 and τ=30 values, which lie *below the persistence floor* of the shipped labels themselves; at τ=30 every condition is worse than persistence. Moving the same head and labels from the published split to a ticker-disjoint split raises test MSE by 2% / 20% / 38% / 36%; the embargoed temporal split costs 2% / 7% / −2% / 24%.

tution. The outcomes span the whole range. **Same-Company-Same-Signal** reproduces: its TSMixer baseline matches the 76 published cells to a median 10^{-5} from the shipped data alone, and its TMLP text model on the authors' OpenAI embeddings matches per-window means within 0.01 and recovers the paper's own finding that embeddings shuffled *within* a company do as well as the real ones. Its training-free baselines on EC and MAEC, run from the authors' notebook, reproduce all 48 published cells at the notebook's three-decimal rounding; with an augmented same-ticker earnings history (outside the benchmark's own training set, all dated before the first test call), Aug_STPEV—no text, no training—lands within 0.05 of KeFVP's published MSE at every horizon on EC and MAEC-15 and below it at $\tau \geq 7$ on MAEC-16. **HTML** does not: the released driver splits features and labels with three independent unseeded shuffles, so it cannot have produced the published table, and with the authors' model classes verbatim on rebuilt inputs the published recipe (ten epochs, learning rate 2×10^{-5}, raw log-volatility targets) is still converging at epoch ten—it beats the paper only at τ=3 on a random split with dropout 0, never reaches persistence at $\tau \geq 15$ on any split, and on the VolTAGE split that every later paper inherited it is worse than persistence at every horizon. The text+audio row is the one HTML cell that does reproduce: with the 27 Praat features concatenated, the shipped-code condition gives 0.658/0.347/0.244/0.165 against the published 0.845/0.349/0.251/0.158—and the same inputs under the paper's chronological split or the VolTAGE split do not, so what reproduces is the random-split protocol the code implements, not the chronological one the paper describes. **Sawhney et al.** cannot run as released (an unshipped call inventory, price files and lexicons, a TensorFlow 2.1 stack), so we port it: the finance SVR alone gives 0.712/0.370/0.218/0.154; the text and audio branches are untrained after the two and one epochs the scripts prescribe, and the published ensemble numbers are reproduced *because* the released code tunes the mixing weights on the test set, where an untrained branch's near-constant output acts as an intercept. **DialogueGAT**'s corpus and labels were never released; ported onto FinCall's speaker turns under the paper's per-year protocol it beats persistence at short horizons and collapses on the 2020 split like every model in §6. **KeFVP**'s released script needed six import-level repairs before it would start (a missing function, a package initialiser importing undefined names, a dropped class header, an unreleased package); its MAEC rows use sentence embeddings regenerated with the authors' own generator because the released ones were never published—and they reproduce: seven of the eight MAEC cells fall within one published standard deviation of the published mean over ten repeats, with matching seed spreads (the eighth, MAEC-16 at τ=15, is the row of their table that does not average to its own stated mean). The EC row waits on the released KePt embedding file *[pending: K1: a 1.2 GB file that only a browser session can download]*. The pattern the substrate audit predicted holds on the field's own code: the results that reproduce are those of price-only or same-company baselines and of a price-series model conditioned on text (KeFVP, whose audio branch is disabled in its own code), and the claims that credit a modality either do not reproduce or reproduce for reasons unrelated to it.

6.9 The reproduced models under our controls

Table 10 answers RQ3 on the field's own code. Each model keeps the configuration of Table 9; only the split changes, and wherever the controlled test set is a subset the unchanged model is scored on exactly those calls. **Removing same-company history costs every text model.** HTML loses 18–82% (text) and 20–70% (text+audio) and falls behind persistence at $\tau \geq 15$; Sawhney's validation-tuned ensemble loses 13–41% and becomes worse than its own finance branch at $\tau \geq 7$; SCSS's TMLP loses 12–27% and is worse in 16 to 18 of 19 rolling quarters. The transcript shuffle of §6, applied to HTML's own trained model, says why: giving each test call the transcript of *another call of the same company* changes its MSE by at most 4% (text; 3% with audio), whereas a call of a *different* company costs 9–23% (16–26% with audio). Two thirds of those same-company substitutes are calls the model trained on; under the ticker-disjoint split none is, the same-company swap still changes nothing (under 1%), and even the different-company swap costs the text model under 5%. What HTML reads from a transcript is which company is speaking. **The price-dominated models do not move**: TSMixer, trained on the same reduced ticker set as TMLP, is mixed around zero, and KeFVP—a

price series conditioned on text, its audio branch disabled upstream—is no worse on either MAEC year (within its repeat spread on MAEC-15, slightly better on MAEC-16). Training size does not explain the split: on EC the ticker-disjoint training sets are as large as the anchors' or larger, on DEC TMLP and TSMixer lose the same third of their training calls, and KeFVP loses a third on MAEC without penalty. **The 30-session embargo matters little** on EC and DEC: HTML and Sawhney, which lose 14–27% of their training calls to it, change by at most 13% in either direction, and SCSS by at most 10%; on MAEC it removes half to two thirds of the training calls, and those rows (ledger) cannot separate leakage from data loss. DialogueGAT's port gives no consistent direction under either control (one seed, 42–60 held-out calls per year). The advantage the text models report is, on their own code and data, largely an advantage of having seen the company.

6.10 Audio by bitrate: the Stage-3 ladder on Earnings25

Earnings25 audio is webcast MP3 at 64 kbps for 281 calls and at 24 kbps or below for 232 (§3.2), so it cannot test "clean" audio; it can test whether the FinCall audio result depends on bitrate *within one quarter*, with the same extractors, heads and controls as Table 5. A single-quarter corpus spans about 105 sessions, so the embargoed temporal split degenerates (five training calls); Table 11 therefore uses the ticker-disjoint split, pooled and per stratum. Audio-plus-past-vol heads beat persistence on Δv (pooled R^2_{OOS} 0.44 at τ=3), but a model given a *random other call's* audio keeps most of that (0.35); the audio increment is 0.09 at τ=3 and at most 0.05 beyond, and per stratum it is small and sign-inconsistent (64 kbps: 0.03 / 0.00 / 0.08 / −0.02; ≤24 kbps: −0.04 / 0.02 / 0.07 / 0.04). With 39 and 55 test calls per stratum these increments are not distinguishable from zero, and the higher-bitrate stratum is not the better one. The FinCall result therefore does not look like an artefact of 40 kbps transcodes: at 64 kbps, on 2025 calls, with a post-cutoff encoder stack, audio is as inert beyond past volatility as it was on FinCall.

6.11 Post-cutoff lookahead

Every head in Tables 2–6 was fit on FinCall's temporal training split; here the same fits are scored, with no retraining and no threshold change, on the 475 Earnings25 calls (held July–December 2025, 97% of them in October–December)—after the training cut-off of every frozen encoder in the stack, and on companies none of the heads has seen. Table 12 shows no post-cutoff degradation for any head that was competitive in-cutoff: HAR-RV rises from 0.46/0.41/0.21/−0.29 to 0.56/0.47/0.32/0.08, and the ridge text and audio heads rise with it. The two segments differ in period and universe (the FinCall test year is 2021, mid-cap-heavy; Earnings25 is S&P 500 in a calm quarter), so absolute levels are not comparable across the cut-off; what is comparable is the *ordering*, and it survives: HAR-RV beats every content head at every horizon on level-v, the content heads track past volatility rather than adding to it (on Δv the frozen text head reaches 0.25 at τ=3 against HAR's 0.56), and at τ=30 only HAR-RV edges past persistence (0.08, CI −0.12 to 0.47). The one head that does degrade is the 3k-dimensional fusion ridge (level-v τ=3: 0.43 to 0.15, CI −0.15 to 0.67), which is unstable rather than contaminated. A lookahead effect would show as content heads that were competitive in-cutoff and collapse post-cutoff; we observe none. This is, to our knowledge, the first frozen-pipeline post-cutoff evaluation in this literature; the ticker fixed-effect GBDT of Stage 1 is absent by construction, since every post-cutoff company is unseen.

Anchor sensitivity of the frozen heads. Table **??** repeats the evaluation with day 0 anchored on the measured EDGAR timestamp on both sides of the cut-off (`result_table_7_measured.csv`; same heads, same splits). HAR-RV barely moves, but the ridge heads that carry a learned past-volatility block change character: their in-cutoff Δv collapse under the after-hours anchor (−1.7 to −3.6 for text, −1.0 to −2.0 for WavLM) disappears under the measured one (+0.04 to +0.34), and the post-cutoff fusion instability shrinks from −7.0 to −0.8 at τ=3. This is consistent with the learned $[v_{\mathrm{pre}}, \mathrm{RV}]$ block absorbing the reaction session that the after-hours rule leaves inside the pre-window for before-open calls, and it identifies the "past-vol ridge is not HAR" gap of the Limitations as at least partly an anchoring artefact. The ordering is unchanged on level-v; on Δv at τ=30 the frozen WavLM+past-vol head is the only one whose post-cutoff CI excludes zero (0.18, CI 0.14–0.24, against HAR's 0.08), a single-quarter, no-DM-test observation we report as exploratory and not as a win. The variant is an input to the re-baselining decision of §3.4, not a change of the primary targets.

7 Discussion

What the evidence shows. On the completed ladder (Stages 0–4), four statements hold. (i) Strong econometric baselines are hard to beat, and the headline metric (R^2_{OOS} vs. persistence) exposes a long-horizon regime failure (FinCall temporal τ=30) that raw MSE would obscure. (ii) Frozen text content does not cleanly add signal over past-volatility and identity baselines on the identity-robust Δv target, and the reason is measurable: call embeddings encode ticker identity (89.5% probe accuracy on FinCall), and within-ticker transcript shuffles leave performance essentially unchanged—an in-corpus replication of the "Same Company, Same Signal" effect (Yu et al., 2025). (iii) Frozen audio content is inert: audio-only heads sit at R^2_{OOS} zero everywhere, the ap-

parent gains of past-vol-augmented audio heads survive a *global* audio shuffle, and WavLM embeddings still encode the speaker's company at 76.3% probe accuracy—identity without predictive content. (iv) Fusion does not rescue the modalities it fuses: multimodal never significantly beats the best unimodal model (RQ2 negative). Together these form one coherent finding: *in this task family, what looks like content signal is overwhelmingly company identity plus past volatility.*

Why this is a benchmark paper. Every one of those statements depends on apparatus the shared substrate does not have. The identity probe and the shuffle controls need a ticker per call, which FinCall-Surprise does not ship; the ticker-disjoint rows need a split the lineage never reports; the regime finding needs per-year breakdowns over recomputed targets rather than inherited label files; and the embargo makes the temporal numbers honest at all (§3.1). A field that evaluates on files with a zero-day embargo, majority company overlap, and an unstated horizon convention cannot distinguish a model from a memory of which company it is looking at. ECVOL-bench is the repair, and the negative results above are what the repaired substrate reports for the modern representations we could run. The reproduction study (§6.7) is the same test applied to the field's own models on the field's own data, and it returns the same diagnosis: the one runnable neural model loses up to 38% MSE once companies are held out, and the strongest published baseline is same-company history.

Engaging the strongest positive claims. The strongest recent positive claim—up to 43.8% of 30-day realised-volatility variance from text+acoustic features (Chen et al., 2025)—reports no identity controls, no released code, and a proprietary corpus. Prosody was the modality most plausibly carrying content orthogonal to identity; our modern audio ladder (WavLM, emotion2vec+, eGeMAPS) finds it inert once past volatility is controlled—albeit on the corpus's 40 kbps source audio (see Limitations and §6.10)—and our shuffle controls predict such gains would not survive a within-ticker (let alone global) shuffle, a check the released suite makes cheap. Likewise, ECC Analyzer (Cao et al., 2024a) reports large MSE reductions from LLM-extracted semantics with closed-API components and no released code; it cannot be reproduced, and we say so in the code-availability audit rather than argue with it.

The pre-registered framing gate. The study was pre-registered with a framing gate: a "positive showcase" spine was to be adopted only if content-bearing models beat the Stage-0/1 floor on Δv or HAR-residual with DM-significant improvements that survived the identity-control suite on at least two of four horizons. The criterion was tested and missed at Stage 2 (text), Stage 3 (audio), and Stage 4 (fusion), and the gate was recorded as resolved toward re-examination in the project's decision log before the benchmark framing was adopted. We report this so that the framing of this paper can be audited against what the data said and when.

What a clean answer would need. Independent of any model, two facts about the corpora limit every result in this literature and are addressed only by new data: the uniform 40 kbps transcode underlying all FinCall-Surprise acoustics, and the absence of any measured call time of day. The first is why the Earnings25 bitrate study (§6.10) matters: at 64 kbps and at $\leq$24 kbps alike the audio increment stays below 0.09 with no consistent sign, so the audio-inert finding does not rest on FinCall's transcode; a lossless corpus would still be the definitive test. The second is why the timestamp retrofit (§3.4) is part of the benchmark rather than a footnote.

8 Conclusion

We audited the evaluation substrate that a decade of multimodal earnings-call volatility prediction shares—inherited label files with a zero-day embargo against 30-day targets, majority company overlap between training and test, no reported ticker-disjoint condition, degenerate label cells, and a horizon convention that downstream work misattributes—and released ECVOL-bench to replace it: reconstructed identity and dates for the one corpus with open audio, targets under both horizon conventions, embargoed and ticker-disjoint splits, strong econometric baselines, and an identity-control suite, all regenerable from one command line. Run over modern open-weight text and audio representations, the benchmark's answer is consistent and negative: frozen text content reads company identity rather than call content (89.5% ticker-probe accuracy; within-ticker shuffles leave performance unchanged); frozen audio content is inert beyond past volatility (its apparent gains survive a global audio shuffle); and fusion never significantly beats the best unimodal model. What the leaderboard literature measures as content signal is, under honest controls, overwhelmingly company identity plus past volatility. The remaining studies close the argument from the outside: the field's released labels sit on a call-date anchor that is one session late for most calls and on a calendar-day convention whose τ=3 window holds a single session for half of them; its best published model loses up to 38% MSE when companies are held out and its best published baseline is same-company history; and nothing here degrades when scored, frozen, on calls held after every encoder's cutoff. The verdict is not that transcripts and audio carry no information about a company's risk, but that the benchmarks this literature uses cannot tell that information apart from knowing which company is speaking—and that, once they can, the past-volatility floor has not been beaten.

Limitations

One post-cutoff quarter. The lookahead (§6.11) rests on 475 calls from a single calm quarter and a different universe from the training corpus, so it bounds contamination in the ordering of models, not in their absolute scores; a second post-cutoff season would tighten it. **Reproductions run on their own data.** Each prior model is re-evaluated on its original dataset, not on ECVOL-bench, to keep "the model fails" separable from "the model does not transfer"; two reproductions carry known costs (KeFVP's MAEC embeddings are unavailable upstream; DialogueGAT stops at τ=15 and requires a corpus rebuild). **MLP-head instability.** The shallow-MLP heads are seed-unstable (across-seed R^2_{OOS} std up to ≈3.4), which is why the consolidated grid uses deterministic ridge/structural canonicals; MLP cells in the appendix should be read as seed means with that caveat. **The past-vol covariate block is not HAR.** The learned $[v_{\mathrm{pre}}, \mathrm{RV}_{1/5/22}]$ ridge used inside Stage-2–4 heads overfits the COVID regime shift on the temporal split (R^2_{OOS} −2.9 at τ=15) where the rigid OLS HAR-RV stays robust (+0.21); the two are labelled distinctly throughout, and the gap is itself a small finding about structural vs. unconstrained conditioning under regime shift. **Short-horizon R^2_{OOS} inflation.** The persistence denominator is noisiest at τ=3 (its MSE is ∼2.8× the τ=7 one), so short-horizon R^2_{OOS} gains are easier to show and should be read accordingly. **Timestamps.** Day 0 rests on the earnings-release time measured from EDGAR (§3.4); 10.7% of FinCall and 31.4% of MAEC calls have no Item 2.02 filing in the window and keep the flagged after-hours fallback, and intraday releases are treated as after-hours. Results under the call-date anchor of prior label sets are not interchangeable with ours (Table 13); the call-date targets are shipped for that comparison. **Regime sensitivity.** FinCall-Surprise spans COVID; the temporal τ=30 baseline failure shows long-horizon level forecasts are regime-sensitive, which is why we emphasise the identity-robust Δv target and report per-year breakdowns. **MAEC coverage.** The 78.6% price-join rate (2015–2018 delistings) makes MAEC a secondary, comparability corpus; it also ships no raw audio, so the audio ladder is FinCall-only. **Source audio quality.** Every FinCall audio file is a uniform 40 kbps MP3 transcode (eight source sample rates; 14 calls at telephone-grade 8 kHz), so aggressive lossy compression underlies all acoustic features. The audio-inert finding is therefore established on compression-damaged audio and may not transfer to studio-quality recordings—a caveat that applies equally to prior positive claims built on the same corpora, and the reason for the Earnings25 re-run. **No diarisation in v1.** Audio uses per-call pooling, not speaker-resolved features; the gender analysis correspondingly uses a dominant-speaker F0 proxy. **Identity reconstruction is imperfect.** 7.1% of FinCall-Surprise calls (4.7% of earnings calls) remain unresolved and are excluded with a reason code; the released override table records every manual decision.

Ethics Statement

Data and licensing. We use only openly licensed corpora (FinCall-Surprise, Apache-2.0; MAEC, CC-BY-SA-4.0) and open-weight models. Derived numeric features inherit and explicitly state their upstream license; raw audio/transcripts from unclear-license sources are never redistributed (*scripts-not-data*). **Gender confound.** Audio features encode speaker gender and female executives are underrepresented in these corpora, which can produce error disparities (Wang et al., 2024; Lin et al., 2024; Mathur et al., 2022); we analyse and report a gender-confound diagnostic rather than silently propagate the bias. **Intended use.** This is a forecasting-research benchmark, not investment advice; predictions are noisy and regime-sensitive, and should not be used for automated trading. **Reproducibility as an ethical commitment.** Given the documented reproducibility crisis in financial NLP (Tatarinov et al., 2025), we pin the environment, checksum every data file, and regenerate all result tables byte-identically from committed artifacts.

References

BAAI. BGE-M3 sentence embeddings. Hugging Face (MIT), 2024. URL https://huggingface.co/BAAI/bge-m3.

Mostapha Benhenda. Look-ahead-bench: a standardized benchmark of look-ahead bias in point-in-time LLMs for finance. arXiv:2601.13770, 2026. URL https://arxiv.org/abs/2601.13770.

Yupeng Cao et al. ECC analyzer: Extracting trading signal from earnings conference calls using large language models for stock volatility prediction. In *Proceedings of the 5th ACM International Conference on AI in Finance (ICAIF)*, 2024a. URL https://arxiv.org/abs/2404.18470.

Yupeng Cao et al. RiskLabs: Predicting financial risk using large language model based on multimodal and multi-sources data. arXiv:2404.07452, 2024b. URL https://arxiv.org/abs/2404.07452.

Yupeng Cao et al. FinAudio: A benchmark for audio large language models in financial applications. arXiv:2503.20990, 2025. URL https://arxiv.org/abs/2503.20990.

Chen et al. Distilling numeral information for volatility forecasting. In *Proceedings of the 30th ACM International Conference on Information and Knowledge Management (CIKM)*, 2021a. doi: 10.1145/3459637.3482089.

Sanyuan Chen et al. WavLM: Large-scale self-supervised pre-training for full stack speech processing. Hugging Face (MIT): microsoft/wavlm-large, 2021b. URL https://huggingface.co/microsoft/wavlm-large.

Xiaoliang Chen et al. The sound of risk: A multimodal physics-informed acoustic model for forecasting market volatility and enhancing market interpretability. arXiv:2508.18653, 2025. URL https://arxiv.org/abs/2508.18653.

Umberto Collodel. Interpreting the interpreter: Can we model post-ECB conferences volatility with LLM agents? arXiv:2508.13635, 2025. URL https://arxiv.org/abs/2508.13635.

Fulvio Corsi. A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 2009. doi: 10.1093/jjfinec/nbp001.

Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 1995. doi: 10.1080/07350015.1995.10524599.

Florian Eyben, Martin Wöllmer, and Björn Schuller. openSMILE / eGeMAPS acoustic feature extraction. github.com/audeering/opensmile-python, 2010. URL https://github.com/audeering/opensmile-python.

Zhenyu Gao, Wenxi Jiang, and Yutong Yan. Detecting lookahead bias in LLM forecasts. arXiv:2512.23847, 2025. URL https://arxiv.org/abs/2512.23847.

Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *The Review of Financial Studies*, 2020. doi: 10.1093/rfs/hhaa009.

Songrun He, Linying Lv, Asaf Manela, and Jimmy Wu. Chronologically consistent large language models. arXiv:2502.21206, 2025. URL https://arxiv.org/abs/2502.21206.

Denglin Jiang, Haoran Zhou, Anshul Wadhawan, Brendan Fahy, Vinay Ramesh, David Weisberg, Dmitriy Derkachevskiy, Helen Sheehan, Srivas Prasad, and Michele Franceschini. Earnings25: A comprehensive 500-hour speech benchmark for finance. Interspeech 2026; arXiv:2607.23813; Zenodo record 10.5281/zenodo.18762168 (CC-BY-4.0), 2026.

Jiazheng Li, Linyi Yang, Barry Smyth, and Ruihai Dong. MAEC: A multimodal aligned earnings conference call dataset for financial risk prediction. In *Proceedings of the 29th ACM International Conference on Information and Knowledge Management (CIKM)*, 2020. doi: 10.1145/3340531.3412879.

Yi-Cheng Lin et al. Emo-bias: A large scale evaluation of social bias on speech emotion recognition. arXiv:2406.05065, 2024. URL https://arxiv.org/abs/2406.05065.

Y. Liu et al. AT-FinGPT: Financial risk prediction via an audio-text large language model. *Finance Research Letters*, 2025. URL https://www.sciencedirect.com/science/article/abs/pii/S1544612325002314.

Alejandro Lopez-Lira, Yuehua Tang, and Mingyin Zhu. The memorization problem: Can we trust LLMs' economic forecasts? arXiv:2504.14765, 2025. URL https://arxiv.org/abs/2504.14765.

Ziyang Ma et al. emotion2vec: Self-supervised pre-training for speech emotion representation. Hugging Face: emotion2vec/emotion2vec_plus_large, 2023. URL https://huggingface.co/emotion2vec/emotion2vec_plus_large.

Puneet Mathur et al. DocFin: Multimodal financial prediction and bias mitigation using semi-structured documents. In *Findings of the Association for Computational Linguistics: EMNLP 2022*, 2022. URL https://aclanthology.org/2022.findings-emnlp.139/.

Hao Niu et al. KeFVP: Knowledge-enhanced financial volatility prediction. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023. URL https://aclanthology.org/2023.findings-emnlp.770/.

ProsusAI. FinBERT: Financial sentiment analysis with bert. Hugging Face, 2019. URL https://huggingface.co/ProsusAI/finbert.

Yu Qin and Yi Yang. What you say and how you say it matters: Predicting stock volatility using verbal and vocal cues. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019. URL https://aclanthology.org/P19-1038/.

Anisha Roy and Dip Roy. MemGuard-Alpha: Detecting and filtering memorization-contaminated signals in LLM-based financial forecasting via membership inference and cross-model disagreement. arXiv:2603.26797, 2026. URL https://arxiv.org/abs/2603.26797.

Yunxin Sang and Yang Bao. DialogueGAT: A graph attention network for financial risk prediction by modeling the dialogues in earnings conference calls. In *Findings of the Association for Computational Linguistics: EMNLP 2022*, 2022. URL https://aclanthology.org/2022.findings-emnlp.117/.

Ramit Sawhney, Piyush Khanna, Arshiya Aggarwal, Taru Jain, Puneet Mathur, and Rajiv Ratn Shah. VolTAGE: Volatility forecasting via text audio fusion with graph convolution networks for earnings calls. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020a. URL https://aclanthology.org/2020.emnlp-main.643/.

Ramit Sawhney, Puneet Mathur, Ayush Mangal, Piyush Khanna, Rajiv Ratn Shah, and Roger Zimmermann. Multimodal multi-task financial risk forecasting. In *Proceedings of the 28th ACM International Conference on Multimedia (MM)*, 2020b. doi: 10.1145/3394171.3413752.

Shi et al. Enhancing volatility forecasting in financial markets: A general numeral attachment dataset for understanding earnings calls. In *Proceedings of IJCNLP-AACL 2023*, 2023. URL https://aclanthology.org/2023.ijcnlp-short.5/.

Dong Shu, Yanguang Liu, Huopu Zhang, and Mengnan Du. FinCall-Surprise: A large scale multi-modal benchmark for earning surprise prediction. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13357–13370, 2026a. URL https://aclanthology.org/2026.acl-long.610/. Data: github.com/Tizzzzy/FinCall-Surprise (Apache-2.0).

Dong Shu, Yanguang Liu, Huopu Zhang, Saisai Hu, Haiyan Zhao, Hekun Huang, and Mengnan Du. MM-FinEval: A multi-task multimodal benchmark for real-world financial forecasting. arXiv:2609.38523, 2026b. URL https://arxiv.org/abs/2609.38523.

Nikita Tatarinov et al. Language modeling for the future of finance: A survey into metrics, tasks, and data opportunities. arXiv:2504.07274, 2025. URL https://arxiv.org/abs/2504.07274.

Wang et al. AMA-LSTM: Pioneering robust and fair financial audio analysis for stock volatility prediction. NAACL 2024 Industry Track; arXiv:2407.18324, 2024. URL https://arxiv.org/abs/2407.18324.

Yutong Yan et al. DatedGPT: Preventing lookahead bias in large language models with time-aware pretraining. arXiv:2603.11838, 2026. URL https://arxiv.org/abs/2603.11838.

Linyi Yang, Tin Lok James Ng, Barry Smyth, and Ruihai Dong. HTML: Hierarchical transformer-based multi-task learning for volatility prediction. In *Proceedings of The Web Conference (WWW)*, 2020.

Linyi Yang et al. NumHTML: Numeric-oriented hierarchical transformer model for multi-task financial forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2022. URL https://arxiv.org/abs/2201.01770.

Ye, Qin, and Xu. Financial risk prediction with multi-round q&a attention network. In *Proceedings of the 29th International Joint Conference on Artificial Intelligence (IJCAI)*, 2020. URL https://www.ijcai.org/proceedings/2020/631.

Ding Yu, Zhuo Liu, and Hangfeng He. Same company, same signal: The role of identity in earnings call transcripts. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 18403–18422, 2025. URL https://aclanthology.org/2025.findings-acl.946/.

Ding Yu, Zhuo Liu, Hao Zhang, and Hangfeng He. Fast numbers, slow language: Bridging quantitative and qualitative earnings signals. arXiv:2606.29734, 2026. URL https://arxiv.org/abs/2606.29734.

A Complete Results

The tables below are generated byte-identically by `ecvol report` from the committed result CSVs (`data/results/`) and are the source of truth for the curated tables in §6; the curated tables copy selected cells from them. They span both corpora (FinCall-Surprise, MAEC), the level-v and Δv targets, both the R^2_{OOS} and MSE metrics, and the temporal and ticker-disjoint splits at all horizons; Result Table 1 carries both horizon conventions (`convention` column), Tables 2–4 the trading-day convention.

A.1 Stage-0/1 baselines (Result Table 1)

Paper / setting	what ran	ours τ=3	7	15	30	τ
HTML (Yang et al., 2020), EC — authors' model classes verbatim, inputs rebuilt (WWM-BERT layer −2), driver corrected						
code: random 70/10/20, dropout 0	text	0.660	0.336	0.253	0.160	1.1
paper: chronological 7:1:2, dropout 0.5	text	0.972	0.439	0.406	0.248	
published split (VolTAGE), dropout 0.5	text	1.231	1.122	1.063	1.479	
persistence on the shipped labels	—	1.485	0.570	0.340	0.202	
code: random 70/10/20, dropout 0	text + 27 Praat	0.658	0.347	0.244	0.165	0.8
paper: chronological 7:1:2, dropout 0.5	text + 27 Praat	0.876	0.520	0.291	0.239	
published split (VolTAGE), dropout 0.5	text + 27 Praat	2.701	1.195	1.436	1.411	
Same-Company-Same-Signal (Yu et al., 2025), DEC — authors' `run.py`, per-window means over 19 rolling quarters						
TSMixer (price history only)	76 cells	0.561	0.252	0.210	0.257	0.5
TMLP on OpenAI embeddings	76 cells	0.608	0.285	0.215	0.202	0.5
TMLP, embeddings shuffled within ticker	76 cells	0.572	0.276	0.213	0.209	0.5
TMLP, embeddings shuffled across all	76 cells	0.646	0.315	0.259	0.249	0.6
Same-Company-Same-Signal, EC / MAEC — authors' notebook, training-free, augmented same-ticker earnings history						
EC, Aug_STPEV	no training	0.569	0.293	0.201	0.122	0.5
MAEC-15, Aug_STPEV	no training	0.443	0.214	0.130	0.112	0.4
MAEC-16, Aug_STPEV	no training	0.470	0.238	0.168	0.113	0.4
KeFVP (Niu et al., 2023) — authors' `final_series_infer.py` (six import-level release defects patched), 10 repeats						
MAEC-15, regenerated BERT-base embeddings	text + price	0.419	0.185	0.123	0.086	0.4
MAEC-16, regenerated BERT-base embeddings	text + price	0.442	0.288	0.363	0.188	0.4
EC, released KePt embeddings	text + price	*[pending: K1: file needs a browser download]*				0.6
Sawhney et al. (2020b), EC — PyTorch port of the released scripts (their labels, split, branches); 5 seeds						
finance SVR (30 past volatilities)	port	0.712	0.370	0.218	0.154	
text BiLSTM, their 2 epochs	port	2.135	1.580	1.384	1.401	
aligned audio, their 1 epoch	port	6.514	5.295	5.368	5.130	
ensemble, (α, β) tuned on test (theirs)	port	0.639	0.298	0.175	0.102	0.6
ensemble, (α, β) tuned on validation	port	0.686	0.324	0.187	0.117	
persistence on their labels	—	1.491	0.563	0.345	0.205	
DialogueGAT (Sang and Bao, 2022) — PyG port on FinCall turns (role-typed speakers); published cells are on their unreleased corpus						
FinCall 2019 (their 2015 protocol)	port	0.694	0.320	0.206	0.134	0.4
persistence	—	1.429	0.566	0.289	0.139	
FinCall 2020 (their 2016 protocol)	port	0.621	0.350	0.392	0.270	0.4
persistence	—	1.409	0.335	0.184	0.101	
FinCall 2021 (their 2017–18 protocol)	port	0.794	0.344	0.182	0.082	0.4
persistence	—	1.301	0.465	0.225	0.118	

Table 9: Reproductions with the authors' own code and data (Result Table 6R-F, T6R.3). Every cell regenerates from one `ecvol reproduce` command; the fidelity ledger lists what is verbatim, patched, substituted or reason-coded per paper. Three of the five reproduce (TSMixer to 10^{-5}; TMLP within single-seed noise, including the paper's real $\approx$ ticker-shuffled finding; KeFVP on MAEC within one published standard deviation on seven of eight cells, with regenerated sentence embeddings). HTML's text-only recipe does not reach its published long-horizon cells on any split, while its text+audio row reproduces at $\tau \geq 7$ under the shipped-code condition (random split, dropout 0) and on no other; Sawhney's published ensemble is reachable with untrained text and audio branches acting as an intercept on the finance SVR once the mixing weights are tuned on the test set, as the released code does.

Model / condition	n_{tr}	n_{te}	τ=3	7	15	30
HTML, authors' classes (EC; seeds 0–2)						
text — shipped-code random split (anchor)	≈390	112	0.660	0.336	0.253	0.160
text — embargoed	285	101	0.726	0.347	0.246	0.178
text — ticker-disjoint	386	109	0.777	0.522	0.393	0.291
persistence on the ticker-disjoint test set	—	109	1.452	0.550	0.327	0.188
text+audio — anchor	≈390	112	0.658	0.347	0.244	0.165
text+audio — embargoed	285	101	0.684	0.345	0.232	0.186
text+audio — ticker-disjoint	386	109	0.789	0.500	0.394	0.281
transcript shuffle, anchor split, the 97 test calls whose company has another call						
text — real transcript	≈390	97	0.683	0.348	0.262	0.165
text — another call of the *same* company	≈390	97	0.693	0.361	0.269	0.169
text — a call of a *different* company	≈390	97	0.745	0.386	0.318	0.203
text+audio — real / same company / different company, τ=3	≈390	97	0.675 / 0.692 / 0.784			
Sawhney et al., port — ensemble with (α, β) tuned on validation (EC; 5 seeds)						
their split (anchor)	328	110	0.686	0.324	0.187	0.117
embargoed	283	100	0.719	0.306	0.184	0.107
ticker-disjoint	383	109	0.775	0.416	0.263	0.161
its own finance SVR, ticker-disjoint	383	109	0.700	0.365	0.256	0.158
Same-Company-Same-Signal, authors' `run.py` (DEC; seed 2021; means over 19 rolling quarters)						
TMLP (text) — paired anchor	900	30	0.591	0.288	0.224	0.211
TMLP (text) — ticker-disjoint	600	30	0.660	0.354	0.280	0.267
TMLP (text) — embargoed (vs anchor 0.608 / 0.285 / 0.215 / 0.202)	895	90	0.585	0.287	0.216	0.208
TSMixer (price) — paired anchor	900	30	0.519	0.241	0.206	0.252
TSMixer (price) — ticker-disjoint	600	30	0.556	0.254	0.194	0.233
persistence on the held-out calls	—	30	1.258	0.457	0.252	0.224
KeFVP, authors' script (MAEC, regenerated BERT-base embeddings; 3 repeats)						
MAEC-15 — paired anchor	535	48	0.470	0.192	0.145	0.114
MAEC-15 — ticker-disjoint	353	48	0.458	0.199	0.150	0.113
MAEC-16 — paired anchor	980	92	0.416	0.248	0.347	0.186
MAEC-16 — ticker-disjoint	661	92	0.378	0.241	0.312	0.178
persistence on the MAEC-16 held-out calls	—	92	0.622	0.265	0.170	0.116

Table 10: The reproduced models under our controls (Result Table 6R-C, T6R.4). Each block keeps the configuration of Table 9 and changes only the split; paired anchors score the unchanged model on exactly the ticker-disjoint test calls. Removing same-company history costs the three text models 12–82% (HTML, Sawhney, TMLP; TMLP is worse in 16–18 of 19 quarters) and leaves both price-dominated models unchanged (TSMixer, KeFVP), although the ticker-disjoint training sets are as large (EC) or cut by the same third (DEC, MAEC) in both groups. The 30-session embargo moves HTML, Sawhney and SCSS by at most 13% in either direction; KeFVP's embargoed MAEC re-splits keep only a half or a third of its training calls, cannot separate leakage from data loss, and are reported in the ledger, not here. DialogueGAT (one seed, 42–60 held-out calls per year) shows no consistent direction and is also left to the ledger.

Stratum	calls (test)		τ=3	τ=7	τ=15	τ=30
All	514 (95)	real	0.435	0.328	0.171	0.220
		global shuffle	0.454	0.327	0.141	0.206
		increment	**−0.019**	**0.001**	**0.030**	**0.014**
64 kbps	281 (39)	real	0.370	0.250	0.109	0.310
		global shuffle	0.383	0.266	0.133	0.303
		increment	**−0.013**	**−0.016**	**−0.024**	**0.007**
≤24 kbps	232 (55)	real	0.343	0.328	0.174	0.145
		global shuffle	0.443	0.354	0.174	0.128
		increment	**−0.100**	**−0.026**	**−0.001**	**0.018**

Table 11: Audio on Earnings25 by bitrate stratum (T9.4; R^2_{OOS} vs. persistence, Δv, ridge on WavLM + past-vol, ticker-disjoint test). The within-ticker shuffle is the identity here (one call per company), so the global shuffle is the control. Call counts are inventory calls with audio at that bitrate (one call at 32 kbps sits in neither stratum); the heads see the 475 admitted, priced calls, and the test count is their ticker-disjoint segment. The audio increment never exceeds 0.03, is negative in 7 of the 12 cells, and has no consistent sign across strata; the 64 kbps stratum is not better than the ≤24 kbps one. Full grid (eGeMAPS, emotion2vec+, fusion, MLP heads, all targets): `result_table_3_earnings25.csv`.

Stage / head	segment	τ=3	τ=7	τ=15	τ=30
0: HAR-RV	in	0.444	0.372	0.158	−0.285
	post	**0.598**	**0.541**	**0.321**	**0.083**
2: ridge text + past-vol	in	0.435	0.339	0.014	−0.454
	post	0.566	0.498	0.259	−0.203
3: ridge WavLM + past-vol	in	0.378	0.245	−0.094	−0.665
	post	0.571	0.431	0.149	−0.323
4: ridge fusion + past-vol	in	0.411	0.317	−0.023	−0.629
	post	0.499	0.462	0.204	−1.927

Table 12: Frozen-pipeline post-cutoff evaluation (T7.2, Result Table 7). Each head is fit once on FinCall's temporal training split and scored unchanged in-cutoff (FinCall test, 2021) and post-cutoff (Earnings25, 475 calls, 2025). No head that was competitive in-cutoff degrades; only the fusion ridge is unstable, at τ=30 (−0.629 in-cutoff, −1.927 post-cutoff). The ordering is preserved out of sample: HAR-RV beats every content head at every horizon, and at τ=30 only HAR-RV edges past persistence, with a CI that includes zero. Month-cluster bootstrap CIs and the Δv rows are in the released CSV.

Head	seg.	τ=3	τ=7	τ=15	τ=30
HAR-RV	in	$0.44 \to 0.46$	$0.37 \to 0.41$	$0.14 \to 0.21$	$-0.29 \to -0.28$
	post	$0.60 \to 0.56$	$0.54 \to 0.51$	$0.35 \to 0.33$	$0.08 \to -0.12$
text + past-vol	in	$0.33 \to -1.71$	$0.28 \to -3.12$	$0.08 \to -1.02$	$-0.08 \to -0.47$
	post	$0.31 \to 0.28$	$0.33 \to 0.27$	$0.23 \to 0.15$	$0.12 \to -0.09$
WavLM + past-vol	in	$0.36 \to 0.26$	$0.26 \to -1.02$	$-0.04 \to -2.01$	$0.05 \to -0.56$
	post	$0.50 \to 0.43$	$0.43 \to 0.32$	$0.15 \to 0.06$	$0.18 \to 0.05$
fusion + past-vol	in	$0.29 \to -1.73$	$0.22 \to -3.13$	$0.04 \to -0.84$	$0.11 \to -0.45$
	post	$-0.83 \to -9.97$	$0.32 \to -2.29$	$0.19 \to 0.02$	$0.01 \to -0.28$

Table 13: Anchor sensitivity: the same frozen heads with day 0 on the call date (the literature's anchor) (Δv, R^2_{OOS} vs. persistence; each cell reads measured → call-date). Ordering on level-v in the CSV.

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
EWMA	0.465*	0.389*	0.259*	0.019
HAR-RV	0.444*	0.372*	0.138*	-0.285*
GARCH(1,1)	0.472*	0.463*	0.298*	0.023
GBDT (ticker FE)	0.353*	0.244*	-0.097	-0.881*

Table 14: fincall — temporal split — level-v — R^2_OOS vs persistence (test) (n=483).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
EWMA	0.520*	0.379*	0.142*	0.112*
HAR-RV	0.570*	0.491*	0.349*	0.345*
GARCH(1,1)	0.561*	0.484*	0.334*	0.298*
GBDT (ticker FE)	0.501*	0.440*	0.292*	0.242*

Table 15: fincall — ticker_disjoint split — level-v — R^2_OOS vs persistence (test) (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
EWMA	0.465*	0.389*	0.259*	0.019
HAR-RV	0.444*	0.372*	0.138*	-0.285*
GARCH(1,1)	0.472*	0.463*	0.298*	0.023
GBDT (ticker FE)	0.340*	0.249*	-0.126	-0.948*

Table 16: fincall — temporal split — Δv — R^2_OOS vs persistence (test) (n=483).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
EWMA	0.520*	0.379*	0.142*	0.112*
HAR-RV	0.570*	0.491*	0.349*	0.345*
GARCH(1,1)	0.561*	0.484*	0.334*	0.298*
GBDT (ticker FE)	0.494*	0.440*	0.292*	0.237*

Table 17: fincall — ticker_disjoint split — Δv — R^2_OOS vs persistence (test) (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
EWMA	0.562*	0.389*	0.223*	0.087*
HAR-RV	0.603*	0.486*	0.321*	0.221*
GARCH(1,1)	0.596*	0.505*	0.325*	0.049
GBDT (ticker FE)	0.563*	0.429*	0.240*	0.113*

Table 18: maec — temporal split — level-v — R^2_OOS vs persistence (test) (n=529).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
EWMA	0.557*	0.388*	0.276*	0.073*
HAR-RV	0.574*	0.447*	0.332*	0.183*
GARCH(1,1)	0.593*	0.481*	0.403*	0.181*
GBDT (ticker FE)	0.483*	0.339*	0.221*	0.091

Table 19: maec — ticker_disjoint split — level-v — R^2_OOS vs persistence (test) (n=532).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
EWMA	0.562*	0.389*	0.223*	0.087*
HAR-RV	0.603*	0.486*	0.321*	0.221*
GARCH(1,1)	0.596*	0.505*	0.325*	0.049
GBDT (ticker FE)	0.562*	0.420*	0.254*	0.150*

Table 20: maec — temporal split — Δv — R^2_OOS vs persistence (test) (n=529).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
EWMA	0.557*	0.388*	0.276*	0.073*
HAR-RV	0.574*	0.447*	0.332*	0.183*
GARCH(1,1)	0.593*	0.481*	0.403*	0.181*
GBDT (ticker FE)	0.477*	0.361*	0.250*	0.095*

Table 21: maec — ticker_disjoint split — Δv — R^2_OOS vs persistence (test) (n=532).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	1.270	0.448	0.211	0.115
EWMA	0.679*	0.274*	0.156*	0.113
HAR-RV	0.706*	0.281*	0.182*	0.148*
GARCH(1,1)	0.671*	0.240*	0.148*	0.112
GBDT (ticker FE)	0.821*	0.338*	0.231	0.216*

Table 22: fincall — temporal split — level-v — MSE (test) (n=483).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	1.368	0.477	0.248	0.262
EWMA	0.657*	0.297*	0.213*	0.232*
HAR-RV	0.589*	0.243*	0.162*	0.171*
GARCH(1,1)	0.602*	0.245*	0.164*	0.184*
GBDT (ticker FE)	0.683*	0.267*	0.176*	0.198*

Table 23: fincall — ticker_disjoint split — level-v — MSE (test) (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	1.270	0.448	0.211	0.115
EWMA	0.679*	0.274*	0.156*	0.113
HAR-RV	0.706*	0.281*	0.182*	0.148*
GARCH(1,1)	0.671*	0.240*	0.148*	0.112
GBDT (ticker FE)	0.838*	0.336*	0.237	0.224*

Table 24: fincall — temporal split — Δv — MSE (test) (n=483).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	1.368	0.477	0.248	0.262
EWMA	0.657*	0.297*	0.213*	0.232*
HAR-RV	0.589*	0.243*	0.162*	0.171*
GARCH(1,1)	0.602*	0.245*	0.164*	0.184*
GBDT (ticker FE)	0.692*	0.267*	0.176*	0.200*

Table 25: fincall — ticker_disjoint split — Δv — MSE (test) (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	1.774	0.712	0.356	0.201
EWMA	0.778*	0.435*	0.276*	0.183*
HAR-RV	0.705*	0.366*	0.242*	0.156*
GARCH(1,1)	0.717*	0.352*	0.240*	0.191
GBDT (ticker FE)	0.775*	0.406*	0.270*	0.178*

Table 26: maec — temporal split — level-v — MSE (test) (n=529).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	1.709	0.632	0.326	0.161
EWMA	0.756*	0.387*	0.236*	0.149*
HAR-RV	0.730*	0.350*	0.218*	0.132*
GARCH(1,1)	0.695*	0.328*	0.194*	0.132*
GBDT (ticker FE)	0.884*	0.417*	0.254*	0.147

Table 27: maec — ticker_disjoint split — level-v — MSE (test) (n=532).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	1.774	0.712	0.356	0.201
EWMA	0.778*	0.435*	0.276*	0.183*
HAR-RV	0.705*	0.366*	0.242*	0.156*
GARCH(1,1)	0.717*	0.352*	0.240*	0.191
GBDT (ticker FE)	0.778*	0.413*	0.265*	0.171*

Table 28: maec — temporal split — Δv — MSE (test) (n=529).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	1.709	0.632	0.326	0.161
EWMA	0.756*	0.387*	0.236*	0.149*
HAR-RV	0.730*	0.350*	0.218*	0.132*
GARCH(1,1)	0.695*	0.328*	0.194*	0.132*
GBDT (ticker FE)	0.894*	0.404*	0.244*	0.146*

Table 29: maec — ticker_disjoint split — Δv — MSE (test) (n=532).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
EWMA	0.404*	0.420*	0.328*	0.145*
HAR-RV	0.419*	0.402*	0.275*	-0.053
GARCH(1,1)	0.388*	0.484*	0.399*	0.145*
GBDT (ticker FE)	0.249*	0.278*	0.105	-0.368*

Table 30: fincall — temporal split — level-v — R^2_OOS vs persistence (test) — calendar-day τ (n=483).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
EWMA	0.470*	0.476*	0.258*	0.036
HAR-RV	0.534*	0.559*	0.418*	0.289*
GARCH(1,1)	0.520*	0.546*	0.409*	0.262*
GBDT (ticker FE)	0.472*	0.498*	0.340*	0.213*

Table 31: fincall — ticker_disjoint split — level-v — R^2_OOS vs persistence (test) — calendar-day τ (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
EWMA	0.404*	0.420*	0.328*	0.145*
HAR-RV	0.419*	0.402*	0.275*	-0.053
GARCH(1,1)	0.388*	0.484*	0.399*	0.145*
GBDT (ticker FE)	0.225*	0.270*	0.101	-0.389*

Table 32: fincall — temporal split — Δv — R^2_OOS vs persistence (test) — calendar-day τ (n=483).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
EWMA	0.470*	0.476*	0.258*	0.036
HAR-RV	0.534*	0.559*	0.418*	0.289*
GARCH(1,1)	0.520*	0.546*	0.409*	0.262*
GBDT (ticker FE)	0.464*	0.503*	0.319*	0.212*

Table 33: fincall — ticker_disjoint split — Δv — R^2_OOS vs persistence (test) — calendar-day τ (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
EWMA	0.566*	0.412*	0.311*	0.169*
HAR-RV	0.562*	0.502*	0.400*	0.265*
GARCH(1,1)	0.579*	0.497*	0.428*	0.203*
GBDT (ticker FE)	0.513*	0.450*	0.357*	0.172*

Table 34: maec — temporal split — level-v — R^2_OOS vs persistence (test) — calendar-day τ (n=529).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
EWMA	0.653*	0.445*	0.337*	0.172*
HAR-RV	0.661*	0.498*	0.396*	0.233*
GARCH(1,1)	0.663*	0.515*	0.452*	0.284*
GBDT (ticker FE)	0.575*	0.416*	0.292*	0.139*

Table 35: maec — ticker_disjoint split — level-v — R^2_OOS vs persistence (test) — calendar-day τ (n=532).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
EWMA	0.566*	0.412*	0.311*	0.169*
HAR-RV	0.562*	0.502*	0.400*	0.265*
GARCH(1,1)	0.579*	0.497*	0.428*	0.203*
GBDT (ticker FE)	0.515*	0.442*	0.370*	0.186*

Table 36: maec — temporal split — Δv — R^2_OOS vs persistence (test) — calendar-day τ (n=529).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
EWMA	0.653*	0.445*	0.337*	0.172*
HAR-RV	0.661*	0.498*	0.396*	0.233*
GARCH(1,1)	0.663*	0.515*	0.452*	0.284*
GBDT (ticker FE)	0.576*	0.420*	0.319*	0.144*

Table 37: maec — ticker_disjoint split — Δv — R^2_OOS vs persistence (test) — calendar-day τ (n=532).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	2.071	0.612	0.298	0.152
EWMA	1.234*	0.355*	0.200*	0.130*
HAR-RV	1.204*	0.366*	0.216*	0.161
GARCH(1,1)	1.271*	0.316*	0.179*	0.130*
GBDT (ticker FE)	1.555*	0.442*	0.267	0.209*

Table 38: fincall — temporal split — level-v — MSE (test) — calendar-day τ (n=483).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	2.550	0.738	0.321	0.220
EWMA	1.352*	0.387*	0.238*	0.212
HAR-RV	1.189*	0.325*	0.186*	0.156*
GARCH(1,1)	1.227*	0.334*	0.188*	0.162*
GBDT (ticker FE)	1.347*	0.371*	0.212*	0.173*

Table 39: fincall — ticker_disjoint split — level-v — MSE (test) — calendar-day τ (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	2.071	0.612	0.298	0.152
EWMA	1.234*	0.355*	0.200*	0.130*
HAR-RV	1.204*	0.366*	0.216*	0.161
GARCH(1,1)	1.271*	0.316*	0.179*	0.130*
GBDT (ticker FE)	1.606*	0.447*	0.268	0.212*

Table 40: fincall — temporal split — Δv — MSE (test) — calendar-day τ (n=483).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	2.550	0.738	0.321	0.220
EWMA	1.352*	0.387*	0.238*	0.212
HAR-RV	1.189*	0.325*	0.186*	0.156*
GARCH(1,1)	1.227*	0.334*	0.188*	0.162*
GBDT (ticker FE)	1.366*	0.367*	0.218*	0.173*

Table 41: fincall — ticker_disjoint split — Δv — MSE (test) — calendar-day τ (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	2.553	0.986	0.492	0.263
EWMA	1.109*	0.579*	0.339*	0.219*
HAR-RV	1.119*	0.491*	0.295*	0.194*
GARCH(1,1)	1.075*	0.496*	0.282*	0.210*
GBDT (ticker FE)	1.243*	0.542*	0.317*	0.218*

Table 42: maec — temporal split — level-v — MSE (test) — calendar-day τ (n=529).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	2.494	0.910	0.426	0.229
EWMA	0.867*	0.505*	0.282*	0.189*
HAR-RV	0.845*	0.457*	0.258*	0.175*
GARCH(1,1)	0.840*	0.441*	0.233*	0.164*
GBDT (ticker FE)	1.059*	0.531*	0.302*	0.197*

Table 43: maec — ticker_disjoint split — level-v — MSE (test) — calendar-day τ (n=532).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	2.553	0.986	0.492	0.263
EWMA	1.109*	0.579*	0.339*	0.219*
HAR-RV	1.119*	0.491*	0.295*	0.194*
GARCH(1,1)	1.075*	0.496*	0.282*	0.210*
GBDT (ticker FE)	1.237*	0.550*	0.310*	0.214*

Table 44: maec — temporal split — Δv — MSE (test) — calendar-day τ (n=529).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	2.494	0.910	0.426	0.229
EWMA	0.867*	0.505*	0.282*	0.189*
HAR-RV	0.845*	0.457*	0.258*	0.175*
GARCH(1,1)	0.840*	0.441*	0.233*	0.164*
GBDT (ticker FE)	1.056*	0.528*	0.290*	0.196*

Table 45: maec — ticker_disjoint split — Δv — MSE (test) — calendar-day τ (n=532).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.430*	0.322*	-0.007	-0.539*
MLP (text)	0.153*	-0.235*	-1.156*	-2.782*
Ridge (past-vol)	0.407*	0.314*	0.079*	-0.382*
MLP (past-vol)	0.439*	0.352*	0.116*	-0.459*
Ridge (text+vol)	0.435*	0.339*	0.034*	-0.454*
MLP (text+vol)	0.219*	-0.115*	-0.918*	-2.504*

Table 46: fincall — temporal split — level-v — R^2_OOS vs persistence (test) (n=483).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.549*	0.449	0.254	0.199
MLP (text)	0.177*	-0.484*	-1.491*	-1.354*
Ridge (past-vol)	0.543	0.467	0.341	0.284
MLP (past-vol)	0.571*	0.512*	0.365*	0.327
Ridge (text+vol)	0.552*	0.464	0.293	0.278
MLP (text+vol)	0.209*	-0.348*	-1.277*	-1.166*

Table 47: fincall — ticker_disjoint split — level-v — R^2_OOS vs persistence (test) (n=484).

A.2 Stage-2 text heads (Result Table 2)

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.182*	0.202	0.044*	-0.133*
MLP (text)	0.013*	0.069*	-0.253	-1.202
Ridge (past-vol)	0.412*	0.314*	0.094*	0.113*
MLP (past-vol)	0.441*	0.330*	0.086*	-0.401*
Ridge (text+vol)	0.333	0.285	0.080*	-0.077*
MLP (text+vol)	0.275	0.222	-0.108	-0.855

Table 48: fincall — temporal split — Δv — R^2_OOS vs persistence (test) (n=483).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.195*	0.217*	0.195*	0.169
MLP (text)	0.046*	-0.024*	-0.108*	-0.061*
Ridge (past-vol)	0.552*	0.469	0.345	0.318*
MLP (past-vol)	0.577*	0.508*	0.381*	0.356*
Ridge (text+vol)	0.399*	0.375	0.256	0.301
MLP (text+vol)	0.423*	0.352*	0.165*	0.224

Table 49: fincall — ticker_disjoint split — Δv — R^2_OOS vs persistence (test) (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.515	0.304*	0.002*	-0.303*
MLP (text)	-0.174*	-1.142*	-2.602*	-5.867*
Ridge (past-vol)	0.575	0.441	0.322*	0.242*
MLP (past-vol)	0.605*	0.470	0.324*	0.240*
Ridge (text+vol)	0.516	0.348*	0.165	0.010*
MLP (text+vol)	-0.116*	-1.036*	-1.853*	-4.494*

Table 50: maec — temporal split — level-v — R^2_OOS vs persistence (test) (n=529).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.556*	0.355	0.116	-0.115*
MLP (text)	0.233*	-0.523*	-1.583*	-3.321*
Ridge (past-vol)	0.530	0.364	0.230	0.159
MLP (past-vol)	0.537	0.389	0.239	0.184
Ridge (text+vol)	0.563*	0.390	0.274	0.133
MLP (text+vol)	0.103*	-0.670*	-1.556*	-3.281*

Table 51: maec — ticker_disjoint split — level-v — R^2_OOS vs persistence (test) (n=532).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.247*	0.148*	0.191	0.098
MLP (text)	0.110*	0.003*	-0.287*	-0.288*
Ridge (past-vol)	0.581	0.444	0.323*	0.242*
MLP (past-vol)	0.615*	0.494*	0.349*	0.258*
Ridge (text+vol)	0.438*	-0.015*	0.233	0.152
MLP (text+vol)	0.407*	0.291*	0.109*	-0.174*

Table 52: maec — temporal split — Δv — R^2_OOS vs persistence (test) (n=529).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.276*	0.267*	0.221	0.113
MLP (text)	0.245*	0.177*	0.070*	-0.112*
Ridge (past-vol)	0.530	0.364	0.230	0.159
MLP (past-vol)	0.559*	0.436*	0.318*	0.182
Ridge (text+vol)	0.453	0.340	0.235	0.093
MLP (text+vol)	0.457	0.322	0.168	-0.070*

Table 53: maec — ticker_disjoint split — Δv — R^2_OOS vs persistence (test) (n=532).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.724*	0.304*	0.212	0.177*
MLP (text)	1.075*	0.553*	0.455*	0.434*
Ridge (past-vol)	0.752*	0.307*	0.194*	0.159*
MLP (past-vol)	0.713*	0.290*	0.186*	0.168*
Ridge (text+vol)	0.718*	0.296*	0.204*	0.167*
MLP (text+vol)	0.992*	0.499*	0.404*	0.402*

Table 54: fincall — temporal split — level-v — MSE (test) (n=483).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.617*	0.263	0.185	0.210
MLP (text)	1.125*	0.708*	0.619*	0.616*
Ridge (past-vol)	0.626	0.254	0.163	0.187
MLP (past-vol)	0.587*	0.233*	0.158*	0.176
Ridge (text+vol)	0.613*	0.256	0.176	0.189
MLP (text+vol)	1.082*	0.643*	0.565*	0.567*

Table 55: fincall — ticker_disjoint split — level-v — MSE (test) (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	1.039*	0.357	0.202*	0.130*
MLP (text)	1.253*	0.417*	0.264	0.253
Ridge (past-vol)	0.746*	0.307*	0.191*	0.102*
MLP (past-vol)	0.710*	0.300*	0.193*	0.161*
Ridge (text+vol)	0.847	0.320	0.194*	0.124*
MLP (text+vol)	0.921	0.348	0.234	0.213

Table 56: fincall — temporal split — Δv — MSE (test) (n=483).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	1.101*	0.374*	0.200*	0.217
MLP (text)	1.306*	0.489*	0.275*	0.278*
Ridge (past-vol)	0.613*	0.254	0.163	0.178*
MLP (past-vol)	0.578*	0.235*	0.154*	0.168*
Ridge (text+vol)	0.822*	0.298	0.185	0.183
MLP (text+vol)	0.790*	0.309*	0.207*	0.203

Table 57: fincall — ticker_disjoint split — Δv — MSE (test) (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.861	0.495*	0.355*	0.261*
MLP (text)	2.082*	1.525*	1.281*	1.377*
Ridge (past-vol)	0.755	0.398	0.241*	0.152*
MLP (past-vol)	0.701*	0.377	0.240*	0.152*
Ridge (text+vol)	0.859	0.464*	0.297	0.198*
MLP (text+vol)	1.981*	1.450*	1.015*	1.102*

Table 58: maec — temporal split — level-v — MSE (test) (n=529).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.758*	0.408	0.288	0.180*
MLP (text)	1.310*	0.962*	0.841*	0.697*
Ridge (past-vol)	0.802	0.402	0.251	0.136
MLP (past-vol)	0.791	0.386	0.248	0.132
Ridge (text+vol)	0.746*	0.385	0.236	0.140
MLP (text+vol)	1.532*	1.055*	0.833*	0.690*

Table 59: maec — ticker_disjoint split — level-v — MSE (test) (n=532).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	1.336*	0.607*	0.288	0.181
MLP (text)	1.578*	0.710*	0.458*	0.258*
Ridge (past-vol)	0.744	0.396	0.241*	0.152*
MLP (past-vol)	0.684*	0.360*	0.231*	0.149*
Ridge (text+vol)	0.997*	0.723*	0.273	0.170
MLP (text+vol)	1.052*	0.505*	0.317*	0.235*

Table 60: maec — temporal split — Δv — MSE (test) (n=529).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	1.237*	0.463*	0.254	0.143
MLP (text)	1.291*	0.520*	0.303*	0.179*
Ridge (past-vol)	0.802	0.402	0.251	0.136
MLP (past-vol)	0.753*	0.357*	0.222*	0.132
Ridge (text+vol)	0.934	0.417	0.249	0.146
MLP (text+vol)	0.929	0.428	0.271	0.172*

Table 61: maec — ticker_disjoint split — Δv — MSE (test) (n=532).

Model	τ=3	τ=7	τ=15	τ=30
ridge eGeMAPS	0.361	0.144*	-0.370*	-1.159*
mlp eGeMAPS	-10.085	-20.164	-38.738	-61.570
ridge eGeMAPS +vol	0.374	0.260	-0.067	-0.677
mlp eGeMAPS +vol	-7.831	-17.515	-37.949	-60.038
ridge WavLM	0.357	0.084*	-0.507*	-1.106*
mlp WavLM	-0.072*	-0.808*	-2.505*	-5.253*
ridge WavLM +vol	0.378	0.245	-0.094	-0.665*
mlp WavLM +vol	0.016*	-0.582*	-1.882*	-4.053*
ridge emotion2vec+	0.354	0.152*	-0.320*	-0.967
mlp emotion2vec+	-0.270*	-1.582*	-3.692*	-6.470*
ridge emotion2vec+ +vol	0.396	0.305*	0.071*	0.398*
mlp emotion2vec+ +vol	-0.313*	-1.580*	-3.559*	-6.632*
ridge WavLM+eGeMAPS	0.354	0.139*	-0.411*	-1.114*
mlp WavLM+eGeMAPS	-1.847	-5.053	-9.675	-14.344
ridge WavLM+eGeMAPS +vol	0.367	0.259	-0.119	-0.763
mlp WavLM+eGeMAPS +vol	-2.005	-5.365	-10.910	-16.691
ridge WavLM+text +vol	0.427*	0.332*	-0.007	-0.527*
mlp WavLM+text +vol	0.213*	-0.050*	-0.823*	-2.165*

Table 62: fincall — temporal split — level-v — R^2_OOS vs persistence (test) (n=483).

Model	τ=3	τ=7	τ=15	τ=
ridge eGeMAPS	0.535	0.320*	0.073*	0.06
mlp eGeMAPS	-7.712	-26.589	-52.220	-53.9
ridge eGeMAPS +vol	0.544	0.426	0.310	0.2
mlp eGeMAPS +vol	-7.149	-21.128	-47.690	-46.8
ridge WavLM	0.514	0.320*	-0.022*	-0.0
mlp WavLM	-0.020*	-0.939*	-2.346*	-2.2
ridge WavLM +vol	0.521	0.401	0.243	0.2
mlp WavLM +vol	-0.074*	-1.072*	-2.573*	-2.3
ridge emotion2vec+	0.533	0.357*	0.085*	0.1
mlp emotion2vec+	-1.783	-4.890	-10.950	-9.6
ridge emotion2vec+ +vol	0.540	0.463	0.335	0.2
mlp emotion2vec+ +vol	-1.860	-5.254	-11.588	-11.1
ridge WavLM+eGeMAPS	0.516	0.305*	-0.071*	-0.0
mlp WavLM+eGeMAPS	-1.442	-4.849	-10.335	-9.3
ridge WavLM+eGeMAPS +vol	0.522	0.389	0.230	0.2
mlp WavLM+eGeMAPS +vol	-1.676	-5.476	-11.476	-10.3
ridge WavLM+text +vol	0.539	0.448	0.276	0.2
mlp WavLM+text +vol	0.228*	-0.288*	-1.033*	-0.86

Table 63: fincall — ticker_disjoint split — level-v — R^2_OOS vs persistence (test) (n=484).

A.3 Stage-3 audio heads (Result Table 3)

FinCall-Surprise only (MAEC ships no raw audio).

Model	τ=3	τ=7	τ=15	τ=
ridge eGeMAPS	0.163*	0.156*	0.059*	0.07
mlp eGeMAPS	-0.096*	-0.004*	-0.228	-0.5
ridge eGeMAPS +vol	0.373	0.268	0.074*	0.10
mlp eGeMAPS +vol	0.319	-0.212	-0.901	-3.6
ridge WavLM	0.156*	0.148*	-0.027	-0.00
mlp WavLM	0.045*	-0.108*	-0.433*	-1.1
ridge WavLM +vol	0.357	0.262	-0.044	0.05
mlp WavLM +vol	0.295	0.087*	-0.387*	-1.27
ridge emotion2vec+	0.155*	0.129*	0.022*	0.02
mlp emotion2vec+	0.006*	-0.029*	-0.337*	-1.0
ridge emotion2vec+ +vol	0.403*	0.307	0.095*	0.06
mlp emotion2vec+ +vol	0.186*	0.107*	-0.218	-0.9
ridge WavLM+eGeMAPS	0.155*	0.145*	0.015*	-0.00
mlp WavLM+eGeMAPS	-0.010*	-0.180*	-0.550*	-1.1
ridge WavLM+eGeMAPS +vol	0.353	0.268	0.033*	0.05
mlp WavLM+eGeMAPS +vol	0.140	-0.014*	-0.581*	-1.49
ridge WavLM+text +vol	0.311	0.237	0.050*	-0.10
mlp WavLM+text +vol	0.323	0.269	-0.103	-0.9

Table 64: fincall — temporal split — Δv — R^2_OOS vs persistence (test) (n=483).

Model	τ=3	τ=7	τ=15	τ=30
ridge eGeMAPS	0.203*	0.119*	0.137*	0.044*
mlp eGeMAPS	0.093*	-0.037*	0.038*	-0.004*
ridge eGeMAPS +vol	0.524	0.431	0.201	0.138*
mlp eGeMAPS +vol	0.328	0.169*	0.179*	0.139*
ridge WavLM	0.178*	0.192*	0.115*	0.025*
mlp WavLM	0.026*	0.011*	-0.140*	0.119*
ridge WavLM +vol	0.456	0.390	0.267	0.232
mlp WavLM +vol	0.424*	0.346*	0.139*	0.050*
ridge emotion2vec+	0.181*	0.185*	0.122*	0.047*
mlp emotion2vec+	0.077*	0.032*	-0.188*	0.210*
ridge emotion2vec+ +vol	0.541*	0.464	0.340	0.317
mlp emotion2vec+ +vol	0.431*	0.323*	0.123*	0.029*
ridge WavLM+eGeMAPS	0.181*	0.194*	0.136*	0.041*
mlp WavLM+eGeMAPS	0.033*	-0.007*	-0.131*	0.119*
ridge WavLM+eGeMAPS +vol	0.442	0.364*	0.248	0.217
mlp WavLM+eGeMAPS +vol	0.327*	0.154*	-0.152*	0.119*
ridge WavLM+text +vol	0.369*	0.348*	0.228	0.201
mlp WavLM+text +vol	0.404*	0.315*	0.128*	0.103*

Table 65: fincall — ticker_disjoint split — Δv — R^2_OOS vs persistence (test) (n=484).

Model	τ=3	τ=7	τ=15	τ=30
ridge eGeMAPS	0.636	0.325*	0.230*	0.246*
mlp eGeMAPS	11.919	13.169	13.212	14.372
ridge eGeMAPS +vol	0.624	0.274	0.171	0.192
mlp eGeMAPS +vol	11.149	10.562	12.088	12.512
ridge WavLM	0.664	0.325*	0.254*	0.266*
mlp WavLM	1.396*	0.926*	0.831*	0.861*
ridge WavLM +vol	0.655	0.286	0.188	0.197
mlp WavLM +vol	1.470*	0.989*	0.887*	0.882*
ridge emotion2vec+	0.639	0.307*	0.227*	0.231*
mlp emotion2vec+	3.808	2.812	2.967	2.799
ridge emotion2vec+ +vol	0.630	0.256	0.165	0.188
mlp emotion2vec+ +vol	3.912	2.985	3.125	3.171
ridge WavLM+eGeMAPS	0.663	0.332*	0.266*	0.270*
mlp WavLM+eGeMAPS	3.340	2.792	2.814	2.700
ridge WavLM+eGeMAPS +vol	0.653	0.291	0.191	0.199
mlp WavLM+eGeMAPS +vol	3.661	3.091	3.097	2.975
ridge WavLM+text +vol	0.631	0.263	0.180	0.191
mlp WavLM+text +vol	1.056*	0.615*	0.505*	0.487*

Table 67: fincall — ticker_disjoint split — level-v — MSE (test) (n=484).

Model	τ=3	τ=7	τ=15	τ=30
ridge eGeMAPS	0.811	0.383*	0.289*	0.248*
mlp eGeMAPS	14.075	9.472	8.379	7.196
ridge eGeMAPS +vol	0.795	0.331	0.225	0.193
mlp eGeMAPS +vol	11.213	8.287	8.213	6.862
ridge WavLM	0.816	0.410*	0.318*	0.242*
mlp WavLM	1.361*	0.809*	0.739*	0.718*
ridge WavLM +vol	0.789	0.338	0.231	0.191*
mlp WavLM +vol	1.250*	0.708*	0.608*	0.580*
ridge emotion2vec+	0.821	0.380*	0.278*	0.226
mlp emotion2vec+	1.612*	1.156*	0.989*	0.858*
ridge emotion2vec+ +vol	0.766	0.311*	0.196*	0.161*
mlp emotion2vec+ +vol	1.667*	1.154*	0.961*	0.877*
ridge WavLM+eGeMAPS	0.820	0.386*	0.298*	0.243*
mlp WavLM+eGeMAPS	3.615	2.709	2.251	1.763
ridge WavLM+eGeMAPS +vol	0.804	0.332	0.236	0.203*
mlp WavLM+eGeMAPS +vol	3.815	2.849	2.511	2.033
ridge WavLM+text +vol	0.727*	0.299*	0.212*	0.175*
mlp WavLM+text +vol	1.000*	0.470*	0.384*	0.364*

Table 66: fincall — temporal split — level-v — MSE (test) (n=483).

Model	τ=3	τ=7	τ=15	τ=30
ridge eGeMAPS	1.062*	0.378*	0.198*	0.106*
mlp eGeMAPS	1.391*	0.449*	0.259	0.180
ridge eGeMAPS +vol	0.797	0.328	0.195*	0.102*
mlp eGeMAPS +vol	0.865	0.543	0.401	0.535
ridge WavLM	1.072*	0.381*	0.217	0.116*
mlp WavLM	1.212*	0.496*	0.302*	0.248
ridge WavLM +vol	0.817	0.330	0.220	0.109*
mlp WavLM +vol	0.895	0.409*	0.292*	0.261*
ridge emotion2vec+	1.074*	0.390*	0.206*	0.112*
mlp emotion2vec+	1.263*	0.461*	0.282*	0.230
ridge emotion2vec+ +vol	0.757*	0.310	0.191*	0.108*
mlp emotion2vec+ +vol	1.033*	0.400*	0.257	0.229
ridge WavLM+eGeMAPS	1.073*	0.383*	0.208*	0.116*
mlp WavLM+eGeMAPS	1.283*	0.528*	0.327*	0.244
ridge WavLM+eGeMAPS +vol	0.822	0.327	0.204*	0.109*
mlp WavLM+eGeMAPS +vol	1.091	0.454*	0.333*	0.286*
ridge WavLM+text +vol	0.875	0.342	0.200*	0.127*
mlp WavLM+text +vol	0.860	0.327	0.233	0.228

Table 68: fincall — temporal split — Δv — MSE (test) (n=483).

Model	τ=3	τ=7	τ=15	τ=30
ridge eGeMAPS	1.091*	0.420*	0.214*	0.250*
mlp eGeMAPS	1.241*	0.495*	0.239*	0.273*
ridge eGeMAPS +vol	0.652	0.272	0.176	0.187
mlp eGeMAPS +vol	0.920	0.397*	0.204*	0.212
ridge WavLM	1.125*	0.386*	0.216*	0.255*
mlp WavLM	1.333*	0.472*	0.283*	0.313*
ridge WavLM +vol	0.745	0.291	0.182	0.204
mlp WavLM +vol	0.788*	0.312*	0.214*	0.246*
ridge emotion2vec+	1.121*	0.389*	0.218*	0.249*
mlp emotion2vec+	1.263*	0.462*	0.295*	0.317*
ridge emotion2vec+ +vol	0.628*	0.256	0.164	0.179
mlp emotion2vec+ +vol	0.778*	0.323*	0.218*	0.254*
ridge WavLM+eGeMAPS	1.120*	0.385*	0.214*	0.233*
mlp WavLM+eGeMAPS	1.323*	0.481*	0.281*	0.312*
ridge WavLM+eGeMAPS +vol	0.764	0.304*	0.188	0.205
mlp WavLM+eGeMAPS +vol	0.921*	0.404*	0.286*	0.313*
ridge WavLM+text +vol	0.863*	0.311*	0.189	0.183
mlp WavLM+text +vol	0.815*	0.327*	0.217*	0.211

Table 69: fincall — ticker_disjoint split — Δv — MSE (test) (n=484).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
HAR-RV	0.444	0.372	0.138	-0.285
GBDT (ticker FE)	0.353	0.244	-0.097	-0.881
Text (BGE+vol)	0.435	0.339	0.034	-0.454
Audio (WavLM+eG+vol)	0.367	0.259	-0.119	-0.763
Fusion (stack)	0.442	0.390	0.164	-0.227

Table 70: fincall — temporal split — level-v — R^2_OOS vs persistence (test) (n=483).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
HAR-RV	0.570	0.491	0.349	0.345
GBDT (ticker FE)	0.501	0.440	0.292	0.242
Text (BGE+vol)	0.552	0.464	0.293	0.278
Audio (WavLM+eG+vol)	0.522	0.389	0.230	0.239
Fusion (stack)	0.556	0.466	0.319	0.365

Table 71: fincall — ticker_disjoint split — level-v — R^2_OOS vs persistence (test) (n=484).

A.4 Consolidated ablation grid (Result Table 4)

One pre-registered canonical model per stage; $*$ marks Holm-corrected significance vs. Stage-1 within the Δv confirmatory family (two-sided—read direction from the value).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
HAR-RV	0.444	0.372	0.138	-0.285
GBDT (ticker FE)	0.340	0.249	-0.126	-0.948
Text (BGE+vol)	0.333	0.285	0.080*	-0.077*
Audio (WavLM+eG+vol)	0.353	0.268	0.033*	0.050*
Fusion (stack)	0.406*	0.347*	0.066*	0.100*

Table 72: fincall — temporal split — Δv — R^2_OOS vs persistence (test) (n=483).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
HAR-RV	0.570	0.491	0.349	0.345
GBDT (ticker FE)	0.494	0.440	0.292	0.237
Text (BGE+vol)	0.399*	0.375	0.256	0.301
Audio (WavLM+eG+vol)	0.442	0.364	0.241	0.217
Fusion (stack)	0.531	0.465	0.340	0.396*

Table 73: fincall — ticker_disjoint split — Δv — R^2_OOS vs persistence (test) (n=484).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
HAR-RV	0.603	0.486	0.321	0.221
GBDT (ticker FE)	0.563	0.429	0.240	0.113
Text (BGE+vol)	0.516	0.348	0.165	0.010
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 74: maec — temporal split — level-v — R^2_OOS vs persistence (test) (n=529).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
HAR-RV	0.574	0.447	0.332	0.183
GBDT (ticker FE)	0.483	0.339	0.221	0.091
Text (BGE+vol)	0.563	0.390	0.274	0.133
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 75: maec — ticker_disjoint split — level-v — R^2_OOS vs persistence (test) (n=532).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
HAR-RV	0.603	0.486	0.321	0.221
GBDT (ticker FE)	0.562	0.420	0.254	0.150
Text (BGE+vol)	0.438*	-0.015*	0.233	0.152
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 76: maec — temporal split — Δv — R^2_OOS vs persistence (test) (n=529).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
HAR-RV	0.574	0.447	0.332	0.183
GBDT (ticker FE)	0.477	0.361	0.250	0.095
Text (BGE+vol)	0.453	0.340	0.235	0.093
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 77: maec — ticker_disjoint split — Δv — R^2_OOS vs persistence (test) (n=532).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	1.270	0.448	0.211	0.115
HAR-RV	0.706	0.281	0.182	0.148
GBDT (ticker FE)	0.821	0.338	0.231	0.216
Text (BGE+vol)	0.718	0.296	0.204	0.163
Audio (WavLM+eG+vol)	0.804	0.332	0.236	0.203
Fusion (stack)	0.708	0.273	0.176	0.141

Table 78: fincall — temporal split — level-v — MSE (test) (n=483).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	1.368	0.477	0.248	0.262
HAR-RV	0.589	0.243	0.162	0.171
GBDT (ticker FE)	0.683	0.267	0.176	0.198
Text (BGE+vol)	0.613	0.256	0.176	0.189
Audio (WavLM+eG+vol)	0.653	0.291	0.191	0.199
Fusion (stack)	0.607	0.255	0.169	0.166

Table 79: fincall — ticker_disjoint split — level-v — MSE (test) (n=484).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	1.270	0.448	0.211	0.115
HAR-RV	0.706	0.281	0.182	0.148
GBDT (ticker FE)	0.838	0.336	0.237	0.224
Text (BGE+vol)	0.847	0.320	0.194*	0.124*
Audio (WavLM+eG+vol)	0.822	0.327	0.204*	0.109*
Fusion (stack)	0.755*	0.292*	0.197*	0.103*

Table 80: fincall — temporal split — Δv — MSE (test) (n=483).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	1.368	0.477	0.248	0.262
HAR-RV	0.589	0.243	0.162	0.171
GBDT (ticker FE)	0.692	0.267	0.176	0.200
Text (BGE+vol)	0.822*	0.298	0.185	0.183
Audio (WavLM+eG+vol)	0.764	0.304	0.188	0.205
Fusion (stack)	0.642	0.255	0.164	0.158*

Table 81: fincall — ticker_disjoint split — Δv — MSE (test) (n=484).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	1.774	0.712	0.356	0.201
HAR-RV	0.705	0.366	0.242	0.156
GBDT (ticker FE)	0.775	0.406	0.270	0.178
Text (BGE+vol)	0.859	0.464	0.297	0.198
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 82: maec — temporal split — level-v — MSE (test) (n=529).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	1.709	0.632	0.326	0.161
HAR-RV	0.730	0.350	0.218	0.132
GBDT (ticker FE)	0.884	0.417	0.254	0.147
Text (BGE+vol)	0.746	0.385	0.236	0.140
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 83: maec — ticker_disjoint split — level-v — MSE (test) (n=532).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	1.774	0.712	0.356	0.201
HAR-RV	0.705	0.366	0.242	0.156
GBDT (ticker FE)	0.778	0.413	0.265	0.171
Text (BGE+vol)	0.997*	0.723*	0.273	0.170
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 84: maec — temporal split — Δv — MSE (test) (n=529).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	1.709	0.632	0.326	0.161
HAR-RV	0.730	0.350	0.218	0.132
GBDT (ticker FE)	0.894	0.404	0.244	0.146
Text (BGE+vol)	0.934	0.417	0.249	0.146
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 85: maec — ticker_disjoint split — Δv — MSE (test) (n=532).

A.5 Substrate audit (Result Table 5R)

The full per-file audit behind Table 1 (file SHA-256 per repository commit, per-split counts, embargo and duplicate checks, sentinel-zero cells per split, binary-column detection per file, and the per-horizon MAEC label diff) is released as `data/results/result_table_5r.csv` and `result_table_5r_labels.csv`, with the lineage mirror manifest under `raw/ref/lineage/`.

A.6 Reproductions and code-availability audit (Result Table 6R)

The complete reproduction grid behind Table 8 is released as `data/results/result_table_6r.csv` (HTML on EC: three seeds, tuned α, persistence per cell) and `result_table_6r_scss.csv` (Same-Company-Same-Signal on DEC: PEV and STPEV per rolling quarter and horizon, 2021–2023, with the published 2023 cells alongside). Table 86 is the code-availability audit.

Paper	venue	probe (date)	verdict	reason
Qin & Yang, What You Say and How You Say It Matters	ACL 2019	200, 0.0 code files, pushed 2022-08-04 (2026-09-09)	partial	dataset released (Driv
Yang et al., HTML	WWW 2020	200, 15.0 code files, pushed 2024-01-05 (2026-09-09)	partial	model classes run ve set_random_seed and dent unseeded shuffle dead Drive links — re
Sawhney et al., multimodal financial forecasting	ACM MM 2020	200, 112.0 code files, pushed 2020-10-26 (2026-09-09)	partial	TF 2.1 / Keras 2.3.1 ventory, price files an record; no multi-task Torch (T6R.3)
Sawhney et al., VolTAGE	EMNLP 2020	200, 5.0 code files, pushed 2021-11-16 (2026-09-09)	partial	ships the EC split/lab embeddings as .pkl; n
Niu et al., KeFVP	EMNLP Findings 2023	200, 42.0 code files, pushed 2024-01-28 (2026-09-09)	partial	requirements.txt does import-level repairs (names, a dropped cla imports); MAEC emb generator); EC KePt p
DialogueGAT	Findings of EMNLP 2022	200, 6.0 code files, pushed 2023-06-12 (2026-09-09)	partial	MIT code, tau<=15 or labels never released onto FinCall turns (T
Yu, Liu & He, Same Company, Same Signal	Findings of ACL 2025	200, 38.0 code files, pushed 2026-08-13 (2026-09-09)	runnable	MIT; ships DEC with mals, TSMixer 76 cel on the authors' embed
Yang et al., FinTrust	ACL 2023 short	200, 4.0 code files, pushed 2024-01-22 (2026-09-09)	partial	perturbation harness volatility model
Sound of Risk	2025	200, 0.0 code files, pushed 2025-08-26 (2026-09-09)	none	repository holds figu corpus
DeFVP	2024	409 (2026-09-09)	none	repository exists but i
ECHO-GL	2024	200, 3.0 code files, pushed 2024-02-26 (2026-09-09)	none	README states the volatility
ECC Analyzer	2024	no_url	none	no code released; GP
RiskLabs	2024	no_url	none	no code released; GP
NumHTML	AAAI 2022	no_url	none	paper promises code;
GNA-Vol		no_url	none	artefact URL returns
AMA-LSTM		no_url	none	10-byte stub repositor
AT-FinGPT		no_url	none	paywalled; no artefac

Table 86: Code-availability audit of the ECC volatility literature (T6R.2/T6R.3; regenerated by `ecvol audit code`, live GitHub probe on the date shown). One of seventeen papers ships code that runs as released; seven ship code with a broken environment, missing data, or defects that must be repaired before it starts (HTML's driver imports TensorFlow-1 symbols and decorrelates features from labels; KeFVP needs six import-level repairs); nine ship nothing usable. The reason column records what the faithful-reproduction sprint (Table 9) did with each.